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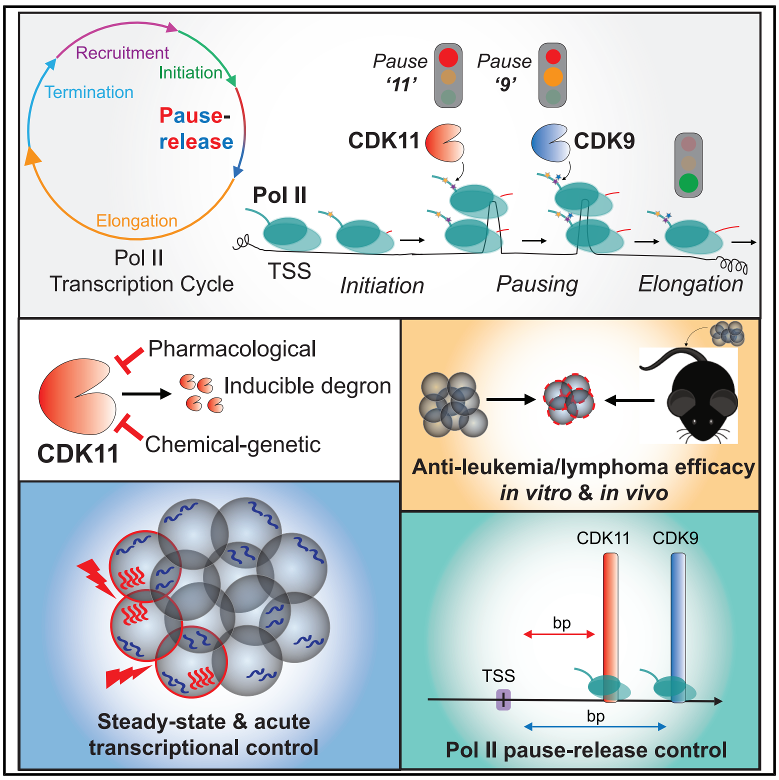
A CDK11-dependent RNA polymerase II pause-checkpoint precedes CDK9-mediated transition to transcriptional elongation

Graphical abstract



Highlights

- CDK11 controls Pol II at a checkpoint upstream of CDK9 within the pause-release zone
- CDK11 inhibition alters Pol II chromatin occupancy and nascent RNA synthesis
- CDK11 is essential for steady-state and induced transcriptional responses
- CDK11 inhibition has therapeutic potential to treat aggressive blood cancers

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In brief

Devlin et al. posit a critical role for CDK11 in Pol II transcription cycles, operating within a dynamic multi-checkpoint pause-release zone to regulate steady-state and inducible gene expression. CDK11 kinase activity facilitates Pol II transition from a paused to an elongative state at a regulatory checkpoint upstream of and distinct from CDK9.

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A CDK11-dependent RNA polymerase II pause-checkpoint precedes CDK9-mediated transition to transcriptional elongation

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SUMMARY

Controlled gene expression is achieved through the intricate regulation of RNA polymerase II (Pol II) progression through transcription-cycle checkpoints. While the contribution of CDK9 for Pol II pause-release is well established, the requirement for other cyclin-dependent kinases (CDKs) has not been fully elucidated. In this study, we propose a critical role for CDK11 in the Pol II pausing-to-elongation transition at a checkpoint that precedes and is independent from CDK9. Selective CDK11 inhibition or degradation results in acute ablation of RNA synthesis near the beginning of transcriptional units and genome-wide stalling of Pol II at transcription start site (TSS)-proximal regions. High-resolution chromatin immunoprecipitation and precision nuclear run-on assays reveal spatial differences between CDK11- and CDK9-dependent Pol II pause sites, with CDK11 regulating Pol II upstream of the CDK9 checkpoint within the pausing zone. Cancer cells exhibit profound reliance on CDK11, with CDK11 inhibition reducing tumor burden in *in vivo* models of blood cancer, demonstrating the importance of CDK11-dependent Pol II regulation for aggressive hematological malignancies.

INTRODUCTION

RNA polymerase II (Pol II) transcription is tightly controlled at multiple levels with discrete molecular checkpoints engaged to precisely regulate gene expression.[1,2] Pol II transcription dysregulation may be considered a universal feature of cancer cells, fueling accelerated proliferation, enhanced survival, aberrant metabolism, and increased invasion/migration.[3–5] Transcriptional cyclin-dependent kinases (tCDKs) have well-defined roles in gene expression control, with different tCDKs and their cognate cyclins (CCNs) modulating Pol II activity throughout distinct phases of the transcription cycle. This includes promoter recruitment (CDK8/CCNC), transcription initiation (CDK7/CCNH), pause-release (CDK9/CCNT1), transcriptional elongation (CDK12/CCNK and CDK13/CCNK), and termination (CDK9/CCNT1) (reviewed in Parua and Fisher[1] and Vervoort et al.[2]). tCDK-directed Pol II regulation is inextricably linked to the Pol II carboxyl-terminal domain (CTD), a repetitive heptad sequence ($Tyr_1Ser_2Pro_3Thr_4Ser_5Pro_6Ser_7$) that is dynamically phosphorylated across different residues/repeats throughout each transcription cycle.[6–9] Additional regulatory functions and phospho-substrates of tCDKs that directly impact Pol II transcriptional activity, as well as co-transcriptional processes such as pre-mRNA splicing and termination-associated cleavage and polyadenylation, continue to be discovered.

Within each Pol II transcription cycle, the pause-release checkpoint is a rate-limiting step that fine-tunes gene expression.[10–15] Typically occurring within approximately 20–60 bp of the transcription start site (TSS), transcriptionally engaged Pol II stalls in an inactivate conformation through interactions with the 5,6-dichloro-1-beta-D-ribofuranosylbenzimidazole (DRB)-sensitivity-inducing factor (DSIF) and negative elongation factor (NELF) complexes.[16–20] CDK9-mediated phosphorylation of the Pol II CTD (Ser_2), as well as SPT5 and NELF-E subunits of the DSIF and NELF complexes, respectively, is critical for the release of Pol II from pausing.[21–25] Aberrant CDK9 recruitment, and the

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resultant enhancement of Pol II pause-release and transcriptional elongation, is a key mechanism driving oncogenic phenotypes in malignancies with *MYC* amplifications or chromosomal rearrangements between the mixed-lineage leukemia (*MLL*r) histone methyltransferase and components of the super elongation complex (SEC). These cancers consequently exhibit high sensitivity to inhibitors of multiple tCDKs and epigenetic regulators.[26–34] Recent studies have highlighted the importance of protein phosphatases such as PP1 and PP2A in opposing CDK9 activity at this checkpoint, with the latter phosphatase recruited by the large, multi-module Integrator complex, which also facilitates pre-mRNA cleavage and premature termination of stalled Pol II complexes.[35–43]

There is broad recognition that the molecular mechanisms governing the transition from pausing to productive elongation are far more complex than CDK9-dependent phosphorylation of Pol II, SPT5, and NELF complexes. Understanding how this checkpoint is so precisely regulated, and what roles are played by an ever-expanding family of regulatory factors, has been an area of intense research focus. Indeed, rather than a single static checkpoint, Pol II pausing can be better thought of as a continuum, with several studies positing multiple pausing sites (e.g., early and late, first and second, high and low frequency) across the promoter-proximal region, as well as in gene bodies, of individual genes.[44–48] The roles of SPT5/DSIF and NELF for the establishment, maintenance, and relief of paused Pol II complexes have been revealed to be highly intricate,[49] with chemical-genetic depletion studies enabling the identification of a putative NELF-dependent "second" pause site downstream of the CDK9-dependent "first" pausing point[47] and demonstrating SPT5-dependent stabilization of paused Pol II complexes.[48,50,51] Studies have proposed the diverse functional contributions of additional regulatory factors for the coordinated transition of Pol II from pausing to elongation, including polymerase-associated factor complex 1 (PAF1c)[52–56]; histone chaperones SPT6 and facilitates chromatin transcription (FACT), which stabilize/remodel chromatin[57–60]; and topoisomerases, which modulate DNA topology and double-strand breaks enriched at Pol II pausing sites.[61–63]

The transcriptional functions of CDK11, which forms active complexes with members of the Cyclin L (CCNL) family, have been less studied compared with other tCDKs. CDK11 is translated from a common mRNA transcript as a ubiquitously expressed 110 kDa isoform (CDK11^{p110}, referred to herein as CDK11), and a 58 kDa C-terminal kinase domain isoform (CDK11^{p58}), which is expressed during the G2-M cell-cycle phase.[64] The recent availability of a selective and orally bioavailable inhibitor, OTS964, has provided new opportunities to fully dissect the molecular contributions of CDK11 to Pol II transcription-cycle control and gene expression more broadly, and has enabled evaluation of CDK11 therapeutic targeting in cancer.[65,66] Early *in vitro* studies established a role for CDK11 kinase activity for catalysis of pre-mRNA splicing, while pharmacological CDK11 inhibition in cells has more recently highlighted its critical role for spliceosome complex formation via SF3B1 phosphorylation.[67–69] CDK11 is required for S and G2-M cell-cycle phase transcriptional programs, including replication-dependent histone (RDH) genes and centromeric alpha satellite loci,[70,71] and has been shown to positively regulate core-binding factor β expression through direct interaction with the CBFβ promoter,[72] as well as to facilitate viral transcript 3′ end processing through interactions with the TREX/THOC complex.[73]

In this study, we propose a critical role for CDK11 in the Pol II transcription cycle, describing a CDK11-dependent transcriptional checkpoint controlling the release of Pol II from pausing to elongation phases. This checkpoint is distinct from, and precedes, the CDK9-controlled pausing checkpoint. We show that CDK11 is essential for Pol II-dependent RNA synthesis under steady-state conditions and for the activation of stimulus-triggered transcriptional responses, and we highlight the reliance of aggressive hematological cancer cells on CDK11 *in vitro* and *in vivo*. Our findings thus demonstrate that cancer cell dependencies on CDK11-regulated transitions in the Pol II transcription cycle can be therapeutically exploited.

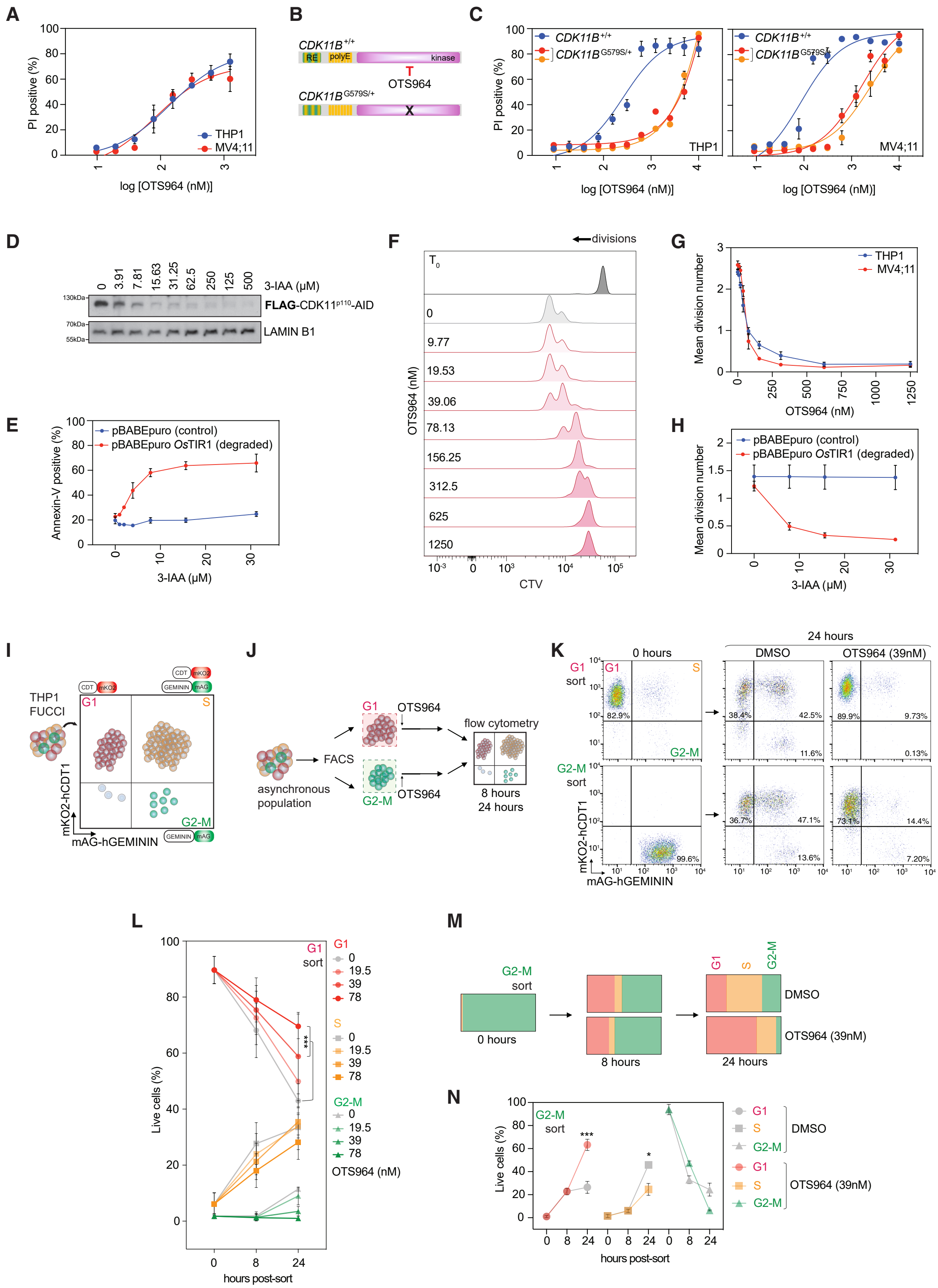
RESULTS

Transcriptional CDK11 is necessary for the survival and proliferation of aggressive AML cells

The dependency of aggressive human blood cancer cells on CDK11 catalytic function and expression was investigated using the small-molecule inhibitor OTS964, as well as two drug-independent chemical-genetic approaches (auxin-inducible degron [AID] and analog-sensitive [AS] mutant kinase) established in *MLL*-rearranged acute myeloid leukemia (*MLL*r-AML) cells with CRISPR-Cas9-mediated knockout of the endogenous *CDK11A*/*B* loci (Figure S1A).

*MLL*r-AML (THP1 and MV4;11), AML with mutant *DNMT3A* and *NPM1* (OCI-AML3), and multiple myeloma cells (MM1.S) exhibited dose-dependent increases in cell death following CDK11 catalytic inhibition with OTS964 (Figures 1A and S1B).[65,66] OTS964 was initially described as a PDZ-binding kinase (PBK) inhibitor[74]; however, publicly available CRISPR-Cas9 data indicate that *PBK* deletion is not detrimental to OTS964-sensitive blood cancer cell lines (THP1, MV4;11, OCI-AML3, MM1.S), in contrast to *CDK11* loss (Figure S1C[75–77]). Critically, leukemia cell killing by OTS964 was ablated in THP1 and MV4;11 cells edited using CRISPR/Cas9 homology-directed repair to harbor a heterozygous *CDK11B* mutation (glycine-579 to serine-579 [G579S]) that confers resistance to this inhibitor[66,67] (Figures 1B, 1C, and S1D). *CDK11B*$^{G579S/+}$ mutant cells did not exhibit cross-resistance to CDK9 (AZ′5576) or CDK12/CDK13 (THZ-531) inhibition, demonstrating the selectivity of this *CDK11B* mutation for OTS964 resistance (Figure S1E).

The necessity of CDK11 for leukemia cell survival was confirmed using rapid *Os*TIR1-dependent degradation of CDK11 protein (Figures 1D and S1F), with dose-dependent increases in cell death measured following 3-indole-acetic acid (3-IAA) treatment of THP1 and MV4;11 cells expressing FLAG-CDK11B-AID (Figures 1E and S1G). This phenotype was recapitulated using a CDK11 AS-mutant kinase, where the ATP-binding pocket gate-keeper methionine is mutated to glycine (M503G), providing sensitivity to the inhibitory ATP-analog 1-NA-PP1 (Figure S1H). THP1 FLAG-CDK11B^{M503G} cells exhibited a dose-dependent increase in cell death following 1-NA-PP1 treatment, while FLAG-CDK11B^{WT} cells exhibited no response (Figure S1I).



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In response to sublethal CDK11 inhibition (<156 nM OTS964), THP1 and MV4;11 cells exhibited reduced proliferation (Figures 1F and 1G), a phenotype recapitulated following CDK11 protein degradation and AS-mutant kinase inhibition (Figures 1H and S1J). The requirement of CDK11 for leukemia cell proliferation was further investigated using the fluorescent-ubiquitination-based cell-cycle indicator (FUCCI) system (Figure 1I). The impact of CDK11 inhibition on THP1-FUCCI cells isolated in either the G1 or the G2-M cell-cycle phase was examined (Figure 1J), with exit from the G1 phase, but not the G2-M phase, significantly impaired by OTS964 (Figures 1K–1N). OTS964-treated G1-isolated cells were prevented from progressing to S and G2-M phases in a dose-dependent manner (Figures 1K and 1L), and while there was minimal impact on the initial transition of G2-M-isolated cells (8 h), OTS964 treatment resulted in significantly higher G1 and significantly lower S phase cell populations at 24 h (Figures 1K, 1M, and 1N). As CDK11 exists as a transcriptional (CDK11^{p110}) and a cell-cycle isoform (CDK11^{p58}), leukemia cell reliance on these different isoforms was investigated. FLAG-CDK11^{p58} over-expression in THP1 cells with CRISPR-Cas9 *CDK11A/B* deletion was unable to significantly rescue cell proliferation, in contrast to full-length FLAG-CDK11^{p110} over-expression (Figures S1K and S1L), suggesting that the CDK11 transcriptional functions are critical for leukemia cell viability.

CDK11 inhibition rapidly decreases hematological tumor burden *in vivo*

The impact of CDK11 kinase inhibition was next assessed *in vivo* using syngeneic mouse models of aggressive *MLL*r-AML and *Myc*-driven B-cell lymphoma, which are characterized by severe splenomegaly, bone marrow and lymph node infiltration of malignant leukemia/lymphoma cells, and highly elevated white blood cell (WBC) populations.[30,31] C57Bl/6 mice bearing murine *Mll-Af9;Nras^{G12}* leukemia[30] received a single maximal tolerated dose of OTS964, and anti-leukemia responses were assessed after 24 h (Figures 2A, S2A, and S2B). CDK11 inhibition significantly reduced splenomegaly compared with vehicle-treated mice (Figures 2B and 2C), accompanied by significant decreases in leukemia cell populations in the spleen and bone marrow (Figures 2D and 2E) and reversal of elevated peripheral blood WBCs (Figure 2F). A murine B-lymphoma model over-expressing transgenic Myc (Eμ-*Myc*[78]) was assessed, with three clonal cell lines exhibiting dose-dependent sensitivity to OTS964-induced cell killing *ex vivo* (Figures S2C and S2D). Similar to *Mll-Af9;Nras^{G12D}*-transplanted mice, C57Bl/6 mice with established Eμ-*Myc* B-lymphoma exhibited significantly reduced splenomegaly, robust decreases in splenic and bone marrow lymphoma cell populations, and normalization of WBC counts at 24 h following a single dose of OTS964 (Figures S2E–S2J). Overall, these results reveal the striking dependency of myeloid- and lymphoid-based blood cancer models with distinct oncogenic drivers on CDK11 kinase activity, indicating that CDK11 inhibition may have the potential to provide robust therapeutic benefits in aggressive hematological cancers.

CDK11 maintains global mRNA synthesis

The transcriptional response of *MLL*r-AML cells to acute perturbation of CDK11 expression and kinase activity was next investigated. Differential gene expression analysis following OTS964 treatment demonstrated transcriptome-wide gene-expression changes, with over 2,000 genes exhibiting significant downregulation, including 1,673 genes shared between THP1 and MV4;11 cells (Figures 3A and 3B). Downregulated genes include oncogenes such as *MYB* and *RUNX1*, and pathway analysis revealed that downregulated genes are associated with critical cellular processes such as mRNA processing, cell growth, protein translation, and metabolism (Figure S3A). Longer-term (>24 h) disruption of CDK11 expression via siRNA-mediated depletion has been linked to the decreased expression of RDH genes.[70] Consistent with this observation, extended catalytic CDK11 inhibition in THP1 cells with OTS964 (24 h) demonstrated decreased expression of multiple RDH genes, with modest yet significant downregulation of these transcripts following acute inhibition of CDK11 (2 h; Figure S3B).

To determine if acute CDK11 disruption impacted global mRNA production, metabolic labeling of nascent RNA using SLAM-seq[79] was performed (Figure 3C). mRNA production in THP1 cells was negatively impacted by acute CDK11 protein degradation, as determined by the significant decrease in "T→C" converted 3′ UTR RNA sequencing (RNA-seq) reads (Figures 3D, S3C, and S3D). Interestingly, despite SLAM-seq assays demonstrating a global decrease in mRNA production in

Figure 1. Hematological cancer cells exhibit a dependency on CDK11

(A) Proportion of PI positive THP1 or MV4;11 cells treated for 48 h. IC_{50}: 142 nM (THP1), 83.10 nM (MV4;11).
(B) Wild-type (*CDK11B*$^{+/+}$) and OTS964-resistant (*CDK11B*$^{G579S/+}$) CDK11.
(C) Proportion of PI positive THP1 or MV4;11 *CDK11B*$^{+/+}$ or *CDK11B*$^{G579S/+}$ cells at 48 h. IC_{50}: 42.1 μM (THP1 *CDK11B*$^{G579S/+}$ clone 1), 28.5 μM (THP1 *CDK11B*$^{G579S/+}$ clone 2), 1.61 μM (MV4;11 *CDK11B*$^{G579S/+}$ clone 1), 2.47 μM (MV4;11 *CDK11B*$^{G579S/+}$ clone 2).
(D) THP1 FLAG-CDK11B-mAID *Os*TIR1 cells treated for 4 h.
(E) Proportion of annexin-V APC-positive THP1 FLAG-CDK11B-mAID cells with/without *Os*TIR1 expression, treated for 48 h.
(F and G) (F) Representative CellTraceViolet (CTV) profiles and (G) mean division number of PI negative *MLL*r-AML cells treated for 48 h.
(H) Mean division number of THP1 FLAG-CDK11B-mAID cells treated for 48 h.
(I) THP1 FUCCI cell cycle indicator system.
(J and K) (J) Schematic and (K) representative profiles of G1/G2-M isolated THP1 FUCCI cells treated for 8 and 24 h.
(L) THP1 FUCCI cells in G1, S, and G2-M phases, treated for 8 and 24 h post G1 isolation.
(M and N) (M) Schematic and (N) quantitation of THP1 FUCCI cells in G1, S, and G2-M phases, treated for 8 and 24 h post G2-M isolation.
(A, C, E, G, H, L, and N) represent the mean ± SEM, and M represents the mean of three independent experiments.
(D, F, and K) are representative of three independent experiments.
(L and N) were analyzed by one-way ANOVA, *$p < 0.05$, ***$p < 0.001$.

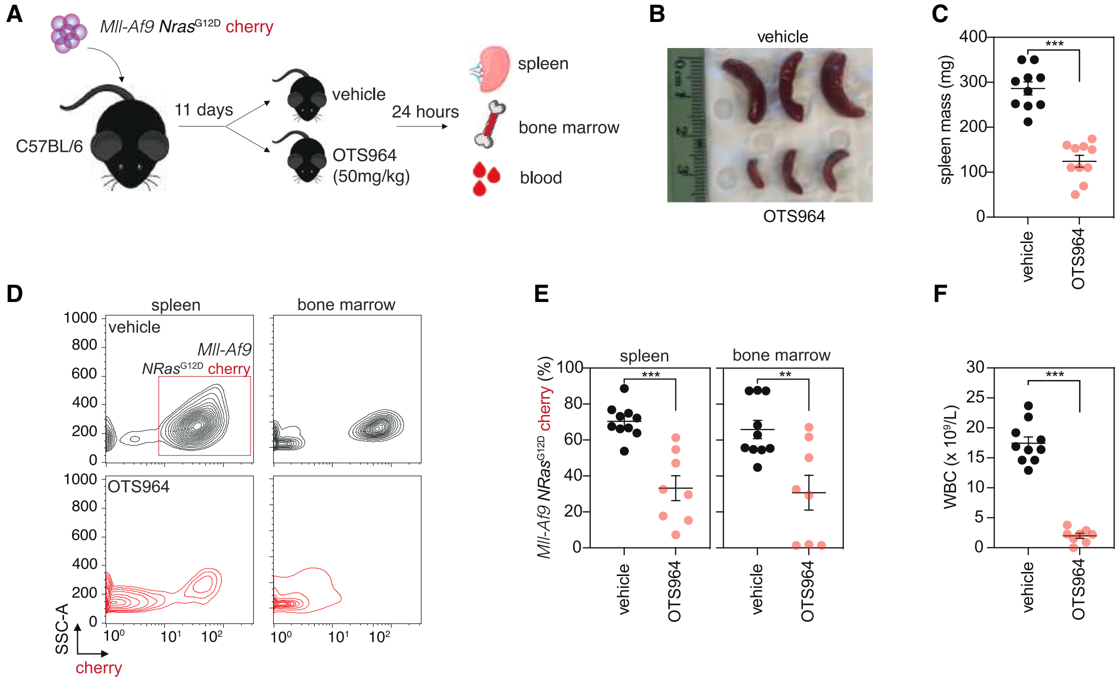

Figure 2. CDK11 inhibition provides therapeutic efficacy against aggressive AML cells *in vivo*
(A) OTS964 therapy in *Mll-Af9* AML syngeneic model.
(B and C) (B) Representative spleen image and (C) spleen mass from *Mll-Af9*-transplanted mice treated for 24 h.
(D) Representative flow cytometry profiles of splenic and bone marrow cell suspensions from *Mll-Af9*-transplanted mice treated for 24 h.
(E) Cherry-positive *Mll-Af9-Nras*$^{G12D/+}$ AML cells in the spleen and bone marrow from *Mll-Af9*-transplanted mice treated for 24 h.
(F) Peripheral blood white blood cell (WBC) concentration in the peripheral blood from *Mll-Af9*-transplanted mice treated for 24 h.
(B and D) is representative of 10 (vehicle) and 8 (OTS964) mice.
(C, E, and F) Each point represents a single mouse; error bars represent SEM. (C,E, and F) were analyzed by one-way ANOVA, $^{**}p < 0.01$, $^{***}p < 0.001$.

response to rapid CDK11 protein loss, between 792 and 1,110 genes showed a significant increase in expression following CDK11 inhibition (Figures 3A and 3B).

Transient-transcriptome sequencing (TT-seq)[80] was next performed to dynamically profile the phase(s) of the transcription cycle at which Pol II-dependent RNA synthesis is modulated by CDK11 (Figure 3E). Acute CDK11 inhibition rapidly ablated RNA synthesis genome-wide within 25 min, with a robust decrease in the average nascent RNA signal across all genes normally expressed in THP1 cells, with a striking decrease at representative loci such as *MYC* (Figures 3F and 3G). RNA synthesis across entire transcriptional units, including gene body and TSS regions, required CDK11 activity (Figure 3H). However, a fold-change comparison between OTS964 and vehicle demonstrated that CDK11 inhibition had the greatest impact at TSS-proximal regions (Figure 3I). By comparing TT-seq profiles across different genomic quadrants discriminated based on gene length, a universal requirement for CDK11 for RNA synthesis was observed at the 5′ end of transcriptional units, with nascent RNA synthesis maintained in the gene body regions of the longest 25%–50% of genes (Q3–Q4) in cells treated with OTS964 (Figures 3J, 3K, and S3E). These data therefore support a key role for CDK11 in maintaining mRNA production and Pol II-dependent RNA synthesis near TSS-proximal regions.

Acute disruption of CDK11 promotes a Pol II pausing-like phenotype

The impact of acute CDK11 inhibition on Pol II chromatin occupancy across transcriptional units was investigated by genome-wide Pol II chromatin immunoprecipitation sequencing (ChIP-seq) in THP1 cells, comparing CDK11 (OTS964) with CDK9 (AZ′5576)[81] inhibition. Increased Pol II abundance at TSS-proximal regions was observed following both CDK11 and CDK9 inhibition, with corresponding decreases in Pol II ChIP-signal across gene bodies (Figures 4A–4C and S4A). A significant decrease in the ratio of chromatin-bound Pol II at gene bodies vs. TSS regions was observed following CDK11 inhibition, phenocopying the effect of CDK9 inhibition (Figures 4D, 4E, and S4B).[25,81] The impact of pharmacological CDK11 kinase inhibition was recapitulated using THP1 FLAG-CDK11B-mAID cells, where Pol II accumulation at TSS-proximal sites, reduced Pol II occupancy across gene bodies, and a significantly lower Pol II gene body/TSS ratio was measured following auxin-induced CDK11 protein degradation (Figures S4C and S4D). Pol II accumulation at the TSS following CDK11 and CDK9 inhibition was also observed using precision nuclear run-on sequencing (PRO-seq[82]; Figure S4E), and detection of genome-wide Pol II stalling at TSS-proximal regions is consistent with the prior observation (Figure 3) that nascent RNA synthesis is acutely ablated near the beginning of transcriptional units following the disruption of CDK11 activity.

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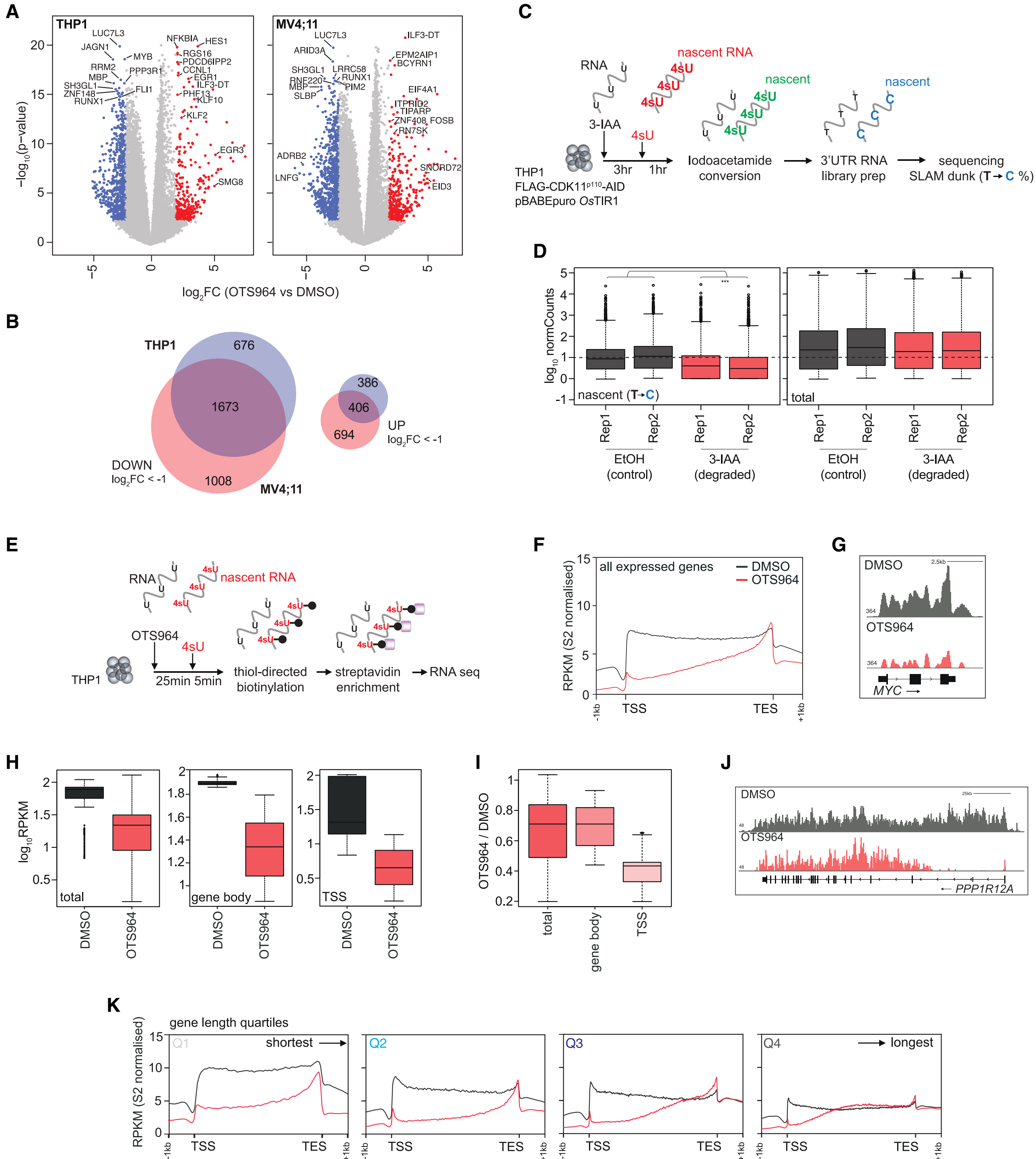

Figure 3. Dynamic transcriptional regulation by CDK11

(A) Volcano plots depicting differentially expressed genes (logFC > 1 | < −1; adj. p value < 0.05) from bulk RNA-seq analysis of THP1 and MV4;11 cells treated with OTS964 (156 nM) for 2 h (n = 14,559).

(B) Overlap of significant (adj. p value < 0.05) downregulated ($log_2FC < -1$) and upregulated ($log_2FC > 1$) genes between THP1 and MV4;11 cells.

(C) SLAM-seq in THP1 FLAG-CDK11B-AID *Os*TIR1 cells treated with 3-IAA (500 μM) for 4 h.

(D) Normalized nascent (T→C converted) and total 3′UTR RNA-seq read counts for THP1 FLAG-CDK11B-AID *Os*TIR1 cells treated with EtOH or 3-IAA for 4 h (n = 46,876 transcripts).

(E) TT-seq of THP1 cells treated with OTS964 (156 nM).

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As Pol II CTD phosphorylation is critical for Pol II progression through transcription-cycle checkpoints, the impact of CDK11 ablation was assessed by reverse phase protein array (RPPA) using Pol II CTD phospho-specific antibodies (Figure S4F). Acute CDK11 kinase inhibition and degradation induced global alterations of Pol II CTD phosphorylation at multiple residues, with robust decreases of Ser_2, Ser_5, and Ser_7 phosphorylation detected (Figures 4F, S4G, and S4H). CDK11 inhibition additionally decreased the abundance of phosphorylated SPT5 (Thr_{806}), another critical phosphorylation event associated with Pol II pause-release (Figures 4F and S4H). Phosphorylation of Pol II CTD residues and SPT5 was significantly rescued in THP1 *CDK11B*$^{G579S/+}$ cells (Figure S4I), while the effects of CDK11 catalytic inhibition on Pol II CTD Ser_2 and SPT5 Thr_{806} phosphorylation were phenocopied in THP1 CDK11 AS-mutant-kinase cells treated with 1-NA-PP1 (Figures 4F and S4J). The *in vivo* biological responses observed in aggressive blood cancer models (Figures 2 and S2) were accompanied by decreased abundance of phosphorylated Pol II (CTD Ser_2, Ser_5, Ser_7) in mice treated with OTS964 (Figure S4K), suggesting a link between reduced tumor burden and OTS964-mediated disruption of CDK11-dependent Pol II transcription. *In vitro* recombinant kinase assays using purified CDK11/CCNL1 complex and human Pol II CTD substrate further demonstrated that, comparable to CDK9 (CDK9/CCNT1), CDK11 has the biochemical capacity to directly phosphorylate Ser_2 residues of the Pol II CTD (Figure 4G), with the *in vitro* catalytic activity of these recombinant kinases efficiently blocked by OTS964 (CDK11) and AZ′5576 (CDK9; Figures S4L–S4N).

The impact of acute CDK11 inhibition, in comparison with CDK9 inhibition, on the Pol II interactome was assessed by immunoprecipitation mass spectrometry (IP-MS). Using an antibody against the RPB1 (POL2RA), the majority of Pol II complex subunits (POLR2A/B/C/D/E/G/H/L) were enriched, as well as other known interactors including the Integrator complex (INTS1/3/4/6), DSIF (SUPT5H), and multiple mRNA splicing/processing proteins (Figures 4H and 4I). 80% (75/94) of these proteins remained significantly enriched in cells treated with OTS964 or AZ′5576 (Figure 4J). Decreased enrichment of 19 proteins was observed following CDK11 (but not CDK9) inhibition, including INTS1 and INTS6, and mRNA splicing/processing factors such as SF3A3 and CNOT1 (Figure 4J). Surprisingly, there was no loss of Pol II interacting proteins following CDK9 inhibition with 56 additional proteins exhibiting significant enrichment, 23 of which were commonly enriched in cells treated with OTS964, with these proteins once again associated with transcriptional and mRNA splicing processes (Figure 4J). These results suggest that CDK11 activity can regulate the Pol II interactome, and this undergoes both similar and distinct changes following CDK11 or CDK9 inhibition.

Together, these results demonstrate that CDK11 regulates Pol II phosphorylation and localization across TSS and gene body regions, with catalytic inhibition or acute degradation of CDK11 inducing a genome-wide redistribution of Pol II chromatin occupancy and pausing-like phenotype similar to CDK9 blockade.

CDK11 occupies TSS-proximal regions in a kinase-dependent manner

The potential recruitment of CDK11 to chromatin where the Pol II pause-to-elongation transition occurs was investigated using ChIP-seq. CDK11 was bound to chromatin genome-wide and was enriched across transcriptional units, with the most dramatic enrichment detected at TSS-proximal regions (Figures 5A, 5B, and S5A). Significantly reduced CDK11 ChIP-signal was observed following CDK11 protein degradation (THP1 FLAG-CDK11-AID), validating the CDK11 antibody used (Figures S5B and S5C).

Analysis of a Pol II ChIP, performed in parallel to the CDK11 ChIP, demonstrated enrichment of Pol II at genomic regions defined as CDK11 peaks, as well as modest enrichment of CDK11 at Pol II peaks, indicating that CDK11 and Pol II can occupy chromatin at the same genomic loci (Figures 5C and S5D). Greater than 40% of annotated CDK11 peaks were within 3 kb of promoter regions, and there was a robust overlap (66.5%) of annotated promoter/proximal-promoter regions that demonstrate enrichment of both CDK11 and Pol II (Figures 5D, 5E, and S5E). CDK11 peaks were also enriched at intronic regions (25.58%), consistent with the previously reported role of CDK11 for pre-mRNA splicing.[67–69] CDK11 occupancy at TSS-proximal regions corresponded with gene expression in exponentially growing THP1 cells, with the CDK11 ChIP signal being most pronounced at TSS-proximal regions of genes with the highest expression levels (top 1,000; Figures 5F and 5G). This trend was also observed for Pol II, with the TSS-associated ChIP-signal increasing from the least (Q1) to the most expressed genes (Q4 and top 1,000; Figures 5F and 5G). Genes with decreased expression following OTS964 treatment exhibited higher CDK11 occupancy compared with upregulated genes, while OTS964 downregulated genes demonstrated greater occupancy of both CDK11 and Pol II at TSS-proximal regions than OTS964 upregulated genes (Figure S5F). CDK11 TSS-proximal occupancy also mirrored Pol II with respect to pausing quadrants, with Pol II and CDK11 TSS coverage being greatest at the most highly paused genes, despite a trend toward decreased overall occupancy of CDK11 and Pol II at genes with higher pausing (Figure S5G).

Interestingly, global CDK11 chromatin binding was dependent on its own kinase activity, with OTS964 treatment resulting in the significant loss of CDK11 occupancy across TSS and gene body regions (Figures 5H and 5I). Importantly, CDK11 occupancy was

(F and G) (F) Average and (G) representative (*MYC*) TT-seq signal in THP1 cells treated with OTS964 for 30 min. Average profiles represent 9,932 expressed genes.
(H and I) (H) $\log_{10}$RPKM coverage for TT-seq signal across entire genes (total, $n = 9{,}932$), gene body and TSS regions, and (I) $\log_2$FC ratio between OTS964 and DMSO.
(J and K) (J) Representative (*PPP1R12A*), and (K) average TT-seq signal profiles across genes subset by gene length (Q1 = shortest 25%; Q4 = longest 25%, $n = 2{,}483$ per quartile).
(A and B) are representative of three independent replicates.
(D) was analyzed by unpaired, two-sided Student's t test ***$p < 0.001$.

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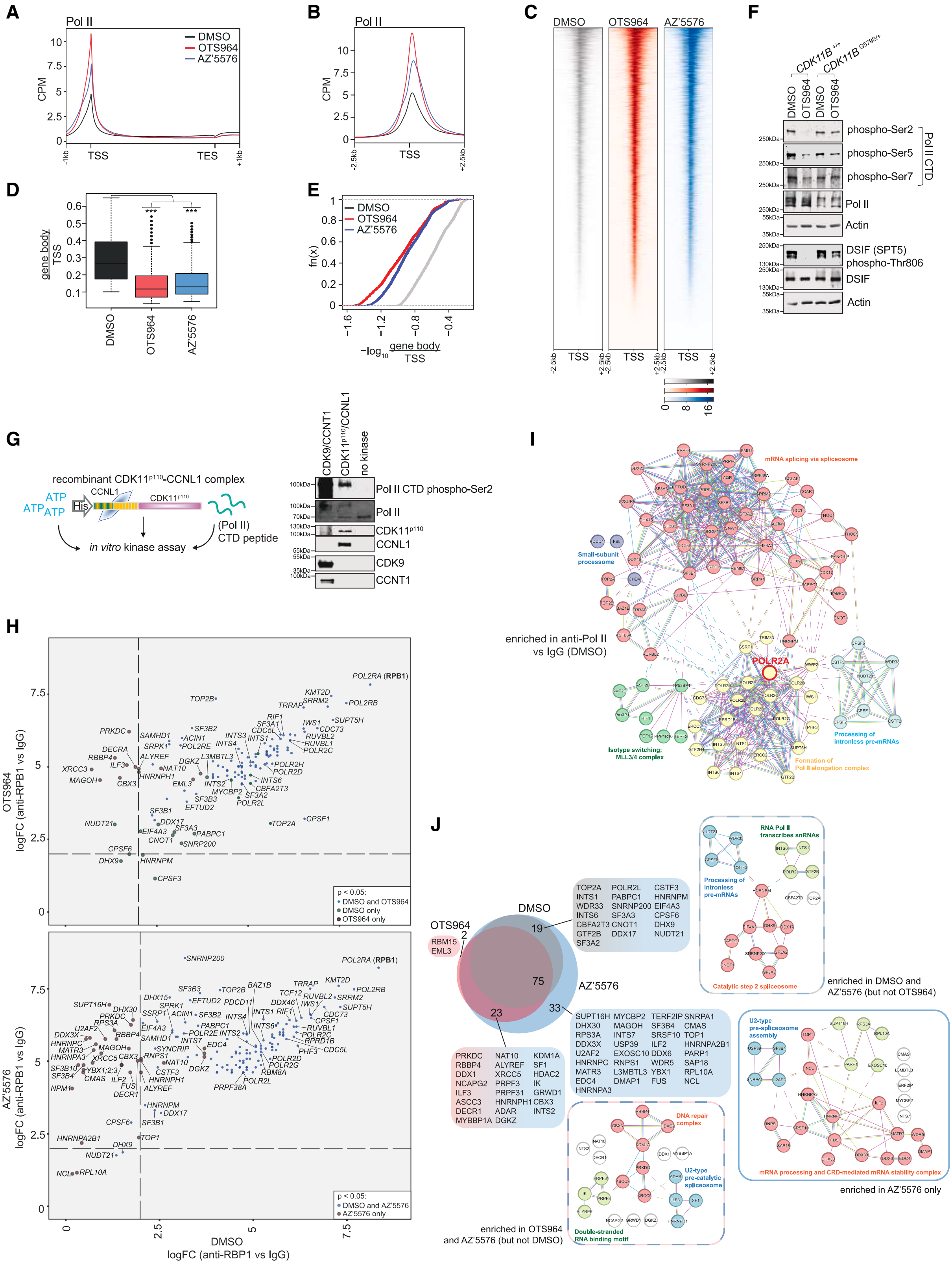

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not impacted by CDK9 inhibition (AZ′5576), indicating that TSS-proximal Pol II accumulation itself is not responsible for CDK11 displacement from chromatin (Figures 5H and 5I). Furthermore, CDK11 inhibition did not alter CDK9 chromatin binding, in contrast to AZ′5576 treatment, which increased CDK9 occupancy at TSS-proximal regions, consistent with previous studies (Figure S5H[81]). IP-MS was performed to profile the CDK11 interactome following targeted kinase inhibition (Figure 5I). The CDK11 interactome included its partner cyclins CCNL1/L2, its previously identified binding partner SAP30BP, which has been shown to be critical for CDK11 catalytic activity,[68,83] and, consistent with previous studies, many splicing factors (Figures 5I and 5J). Additionally, two subunits of the Pol II complex, specifically the CTD-containing subunit POLR2A (RPB1) and POLR2E, were significantly enriched with CDK11, suggesting that CDK11 may be capable of physically interacting with the Pol II complex and associated transcriptional machinery (Figures 5I and 5J). Indeed, there was significant overlap between the Pol II and CDK11 interactomes (45 proteins, 48% and 37%, respectively), with the majority of these proteins associated with mRNA splicing. There was a limited impact of targeted kinase inhibition on the CDK11 interactome, with 94% (115/122) of proteins still enriched following OTS964 treatment (Figures 5I–5K), suggesting that transcriptional impacts of CDK11 inhibition are not linked to broad changes in CDK11 protein-protein interactions. Interestingly, however, both Pol II subunits were no longer enriched with CDK11, a result that is consistent with dramatically reduced CDK11 chromatin occupancy at sites of Pol II accumulation following its inhibition (Figure 5H). Co-immunoprecipitation (co-IP) experiments additionally demonstrated an interaction between CDK11 and Pol II (RBP1) that was robustly perturbed by OTS964 (Figure 5L), supporting a kinase-dependent association between CDK11 and the Pol II complex in cells that is potentially critical for CDK11-dependent regulation of Pol II transcription.

Coupled with the observation that CDK11 inhibition induces Pol II accumulation and ablates RNA synthesis at TSS regions, these results suggest that CDK11 association with Pol II at chromatin, dependent on its own kinase activity, promotes the transition of Pol II from the pausing to the elongation phases of transcription cycles.

CDK11 is required for acute transcriptional elongation triggered by external stimuli

In addition to steady-state gene expression, a role for CDK11 for acute activation of selective transcriptional responses following external stimuli was investigated. CDK11 inhibition by OTS964 significantly antagonized the upregulation of key response genes in THP1 cells treated with bacterial lipopolysaccharide (LPS; Figures 6A, 6B, and S6A). This phenotype was recapitulated using the CDK11 AS-mutant kinase system, where 1-NA-PP1 significantly decreased LPS-induced upregulation of the target genes *CCL3* and *SOD2* in THP1 FLAG-CDK11^{M503G} but not in wild-type FLAG-CDK11 cells (Figure S6B). A broader role of CDK11 in acute transcriptional responses was demonstrated, with OTS964 treatment significantly antagonizing the upregulation of heat-shock response genes (including *HSP27*) in HCT-116 colorectal cancer cells, as well as *CCND1* and *ODC1* following serum re-feeding of G1 cell-cycle-phase-arrested hTERT-immortalized fibroblasts (Figures 6C, 6D, and S6C–S6F). Increased global RNA synthesis triggered by serum re-feeding of G1-arrested HCT-116 cells was also significantly antagonized by CDK11 inhibition (Figures S6G–S6K).

To assess the molecular basis of the CDK11-dependent contribution to acute transcriptional responses, ChIP-seq for Pol II and CDK11 was performed in THP1 cells stimulated with LPS (Figures 6E–6H). Enhanced Pol II recruitment to TSS regions of LPS response genes was observed in response to LPS stimulation, independent of OTS964; however, gene body Pol II occupancy at these loci was significantly decreased following CDK11 inhibition (Figures 6E, 6F, and 6H). Additionally, at LPS-response genes, the gene body/TSS ratio for the Pol II ChIP-signal was significantly lower in cells co-treated with LPS and OTS964 compared with LPS alone (Figure 6G). The CDK11 ChIP-seq showed that CDK11 abundance robustly increased at LPS response genes following stimulation; however, this was dependent on the CDK11 catalytic activity and was blocked by OTS964 (Figures 6E–6H). Overall, these results indicate that while CDK11 kinase activity does not appear to be necessary for Pol II recruitment to key response genes following stimulation, it is important for Pol II transition beyond the TSS-proximal regions into the gene body and subsequent expression of response genes.

CDK11- and CDK9-dependent Pol II pausing phenotypes are distinct and independent

As the data thus far showed generally comparable Pol II pausing-like phenotypes induced by CDK11 and CDK9 inhibition, the contributions of these two kinases to Pol II chromatin occupancy and sites of catalytic activity were next assessed at greater resolution using complementary antibody-DNA and RNA metabolic labeling approaches. Pol II ChIP-nexus[84] enabled quantitative

Figure 4. CDK11 controls global Pol II chromatin occupancy and CTD phosphorylation

(A–C) (A) Pol II ChIP average gene and (B) TSS profiles and (C) TSS metagene occupancy profiles in THP1 cells treated with DMSO, OTS964 (156 nM), and AZ′5576 (170 nM) for 2 h (n = 9,932 expressed genes).

(D and E) (D) Ratio of Pol II ChIP $\log_{10}$CPM gene body vs. TSS coverage, and (E) cumulative distribution of Pol II pausing index.

(F) THP1 and THP1 *CDK11B*$^{G579S/+}$ cells treated with OTS964 (156 nM) for 2 h.

(G) *In vitro* recombinant CDK11/CCNL1 or CDK9/CCNT1 and Pol II CTD kinase assay.

(H) LogFC comparisons of significantly enriched proteins (Pol II vs. IgG, $p < 0.05$) from THP1 cells treated with DMSO and OTS964 (156 nM) or AZ′5576 (170 nM) for 2 h.

(I) STRING protein-protein interaction network of significantly enriched proteins (Pol II vs. IgG, $p < 0.05$, DMSO condition).

(J) Overlap and STRING protein-protein interaction networks of Pol II IP-MS-enriched proteins from cells treated with DMSO, OTS964, and AZ′5576.

(D) was analyzed by unpaired, two-sided Student's t test ***$p < 0.001$. Western blots are representative of three independent experiments.

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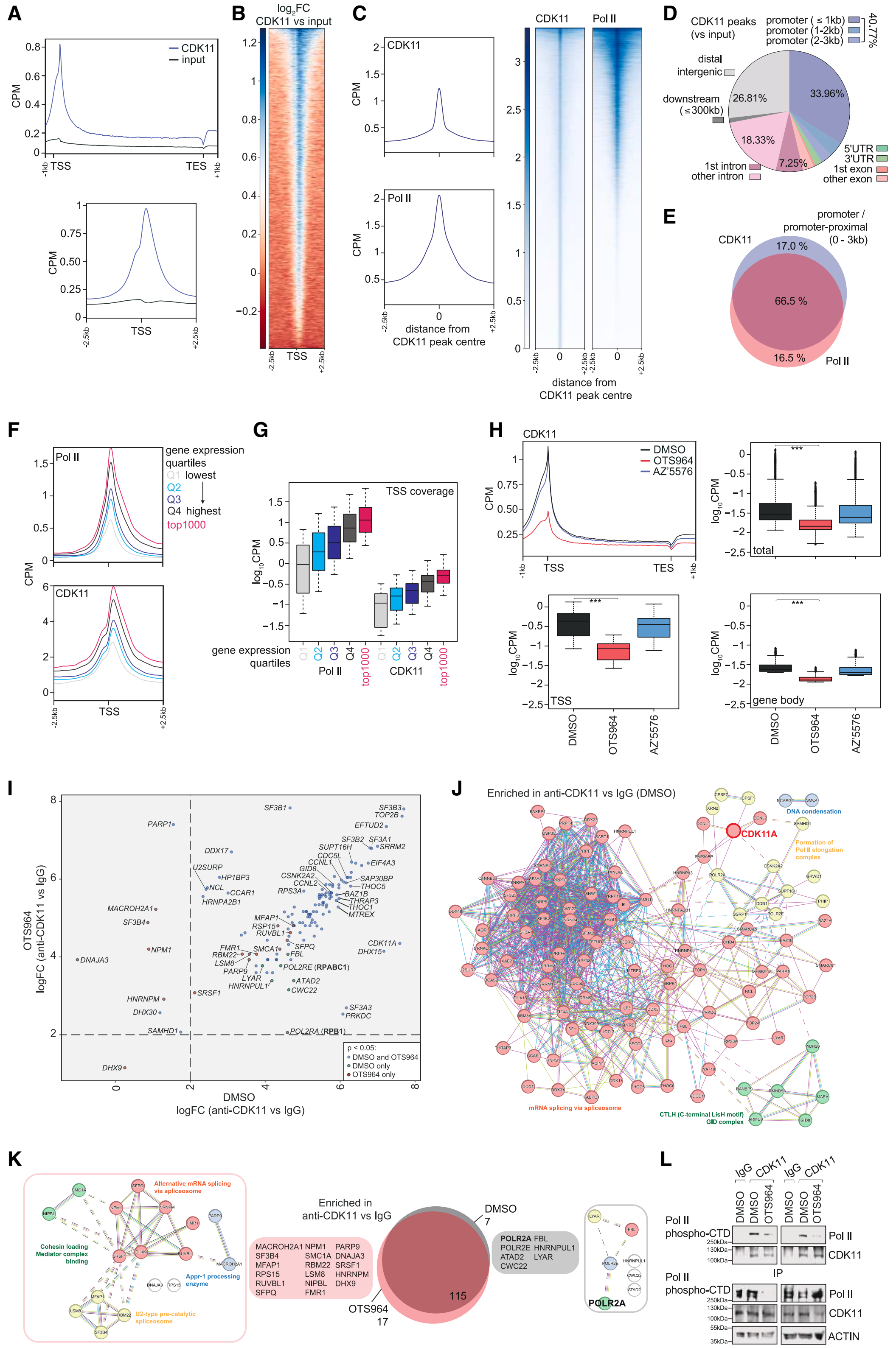

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comparison of the distances between the TSS and the Pol II peak summits (pausing sites) across different genes. There was a demonstrable and significant difference in the median distance between the TSS and OTS964-induced Pol II summits (44.9 bp) compared with the distance between the TSS and AZ′5576-induced Pol II summits (52.6 bp), with CDK11 inhibition globally inducing Pol II accumulation closer to the TSS than CDK9 (Figures 7A–7C and S7A–S7C). This was also observed at representative genes (*IL6R*, *CDC42SE1*, and *FDPS*), where the Pol II ChIP-nexus signal was enriched closer to the annotated TSS in OTS964 vs. AZ′5576 conditions (Figures 7C and 7D). Intriguingly, while there was a clear increase in TSS-proximal Pol II abundance following CDK11 inhibition, no significant difference in the distance between the TSS and Pol II summit was observed between OTS964 and dimethyl sulfoxide (DMSO) conditions (Figures 7A, S7B, and S7C). Additional analysis of PRO-seq data (Figures S7D and S4E) further demonstrates a significant difference in the genomic location of transcriptionally active Pol II in cells treated with OTS964 (111 bp, median distance from TSS) compared with AZ′5576 (136 bp). These DNA- and RNA-based assays thus raise the prospect that CDK11 may be operating upstream of the CDK9 checkpoint to regulate Pol II transcriptional progress.

To further dissect CDK11- vs. CDK9-dependent regulation of Pol II at TSS-proximal regions, PRO-seq assays were performed on THP1 cells that had undergone synchronized Pol II pausing at the CDK9-regulated checkpoint using DRB, prior to Pol II release in the presence or absence of OTS964 or AZ′5576 (Figures 7E and 7F). PRO-seq signal profiles, including at an exemplar gene (*PPP1R12A*), demonstrated robust reactivation of transcription and Pol II progression into gene body regions following DRB-washout in cells released in the presence of DMSO or OTS964 (Figures 7F–7I). However, transcription reactivation was restricted by AZ′5576 treatment, as evidenced by lower PRO-seq signal across gene body regions and a lower gene body/TSS ratio compared with DMSO and OTS964. This suggests that Pol II transcription progression through and beyond the CDK9-dependent/DRB-sensitive pausing checkpoint occurs downstream and independently of CDK11 kinase activity. Additionally, a robust increase in the II-0 (active) hyperphosphorylated Pol II variant[85,86] observed following a 10-min release post-DRB washout was blocked by CDK9 inhibition but was not impaired by inhibition of CDK11 or CDK7 (YKL-5-124), which regulates Pol II initiation (Figure 7J). Importantly, after an extended release (2 h), when new rounds of Pol II transcription would have commenced, there was a clear loss of the II-0 Pol II variant in cells treated with CDK7, CDK11, and CDK9 inhibitors.

To ensure that the phenotype observed was not a consequence of insufficient CDK11 inhibition by OTS964 during the short (10 min) release, cells with DRB-synchronized Pol II pausing were pre-treated with OTS964 for 1 h prior to washout/release. Consistent with the previous result, robust transcriptional reactivation occurred in DMSO- and OTS964-treated cells while AZ′5576 impaired Pol II release into gene bodies (Figure S7E). OTS964 pre-treatment of DRB-synchronized cells also did not negatively impact the re-emergence of the II-0 Pol II variant (Figure S7F). Finally, transcriptional reactivation following Pol II release from the CDK11-dependent checkpoint was assessed, where Pol II blockade was induced by extended OTS964 treatment. Following OTS964 washout, transcriptional reactivation was robustly impaired by continued CDK11 inhibition and, albeit to a lesser extent, by CDK9 inhibition (Figure S7G).

Together, these results support a proposed model whereby CDK11 may modulate Pol II prior to the CDK9-dependent pausing checkpoint and highlight a potential new and distinct regulatory role of CDK11 in the control of Pol II transcription cycles (Figure 7J).

DISCUSSION

This study represents an exciting step forward in our fundamental understanding of how gene expression is controlled, with these findings suggesting that yet another critical layer of regulatory complexity may exist within Pol II transcription cycles. We herein propose a crucial role for CDK11 in the control of a distinct 5′ regulatory checkpoint, placing CDK11 within the Pol II transcription cycle and demonstrating that its kinase activity, upstream of CDK9, is required for Pol II transition from a paused-like to elongative state (Figure 7J). CDK11- and CDK9-dependent pausing phenotypes are spatially distinct, with CDK11 pause sites located significantly closer to the TSS than DRB-sensitive CDK9 pause sites. The phosphorylation status of canonical CDK9 substrates associated with paused-to-elongating Pol II states, specifically the Pol II CTD Ser_2 and SPT5 Thr_{806} residues, is dependent on CDK11 kinase activity. While

Figure 5. Kinase-dependent chromatin localization of CDK11 to TSS-proximal regions

(A and B) (A) CDK11 ChIP average gene and TSS profiles, and (B) TSS metagene occupancy profile (log_2FC vs. input) for THP1 cells (n = 9,932 expressed genes).

(C) Average and metagene occupancy profiles for CDK11 and Pol II ChIP signals at MACS2-called CDK11 peaks (n = 69,450).

(D) Distribution of annotated MACS2-determined CDK11-ChIP peaks (n = 69,096).

(E) Overlap of annotated promoter/promoter-proximal CDK11 and Pol II peaks (n = 15,172).

(F and G) (F) Average TSS profiles and (G) log_{10}CPM coverage for Pol II and CDK11 ChIP signals between gene expression quadrants (n = 2,483 per quartile or 1,000 highest-expressed genes).

(H) CDK11 ChIP average gene profiles and log_{10}CPM coverage for THP1 cells treated with DMSO, OTS964 (156 nM), and AZ′5576 (170 nM) for 2 h (n = 9,932 expressed genes).

(I) LogFC comparisons of significantly enriched proteins (CDK11 vs. IgG, $p < 0.05$) from THP1 cells treated with DMSO or OTS964 (156 nM) for 2 h.

(J) STRING protein-protein interaction network of significantly enriched proteins (CDK11 vs. IgG, $p < 0.05$, DMSO condition).

(K) Overlap and STRING protein-protein interaction networks of CDK11 IP-MS enriched proteins for DMSO and OTS964 conditions.

(L) IgG or CDK11 immunoprecipitation from THP1 cells treated with DMSO or OTS964 (156 nM) for 2 h.

(L) is representative of four independent experiments.

(H) was analyzed by unpaired, two-sided Student's t test ***$p < 0.001$.

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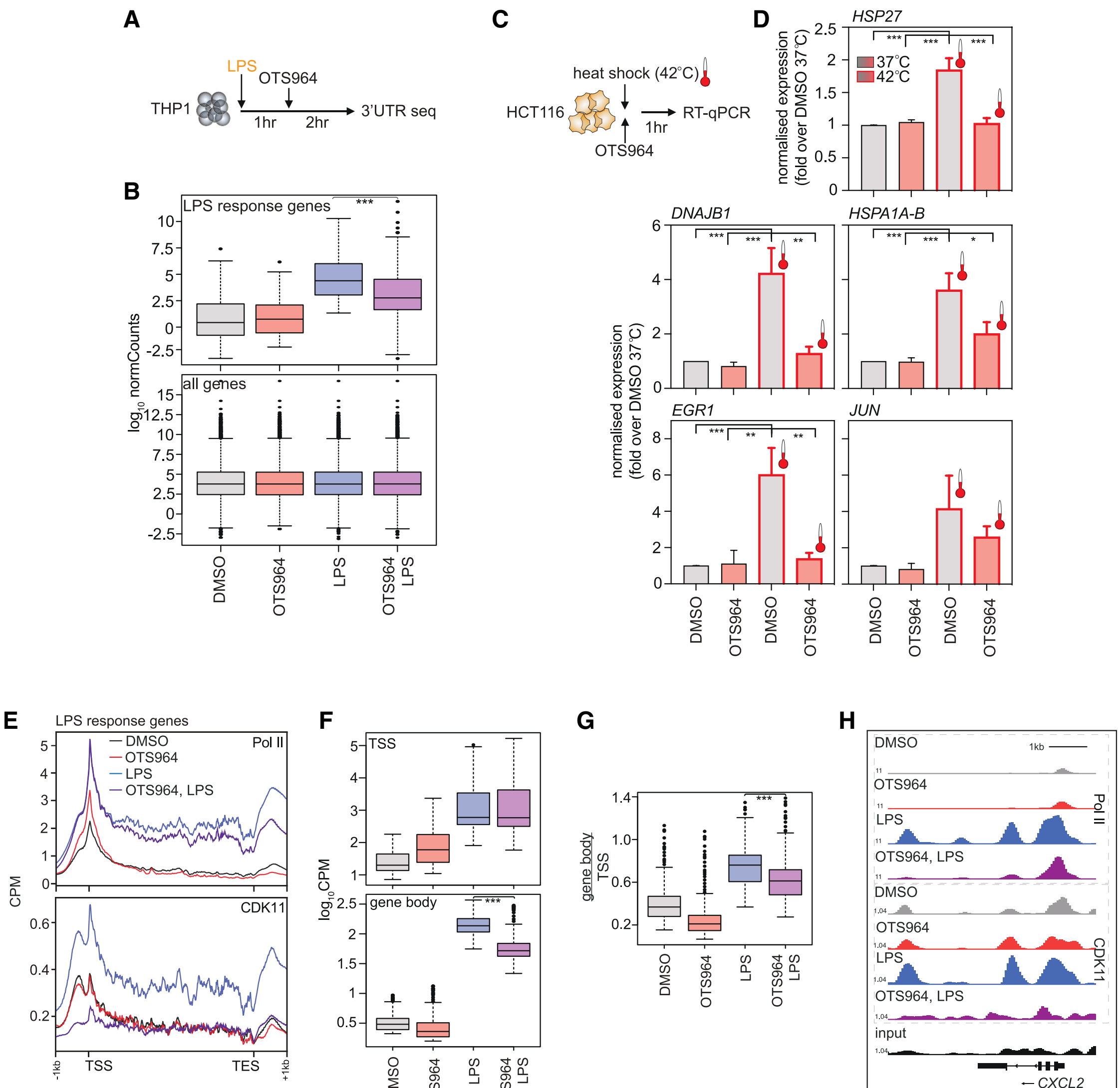


Figure 6. CDK11 is recruited to and promotes the transcription of genes in response to external stimulation

(A) LPS stimulation of THP1 cells.

(B) Normalized bulk RNA-seq read counts for LPS response genes in THP1 cells treated with LPS (1 μg/mL) for 3 h and/or OTS964 (156 nM) for 2 h.

(C and D) (C) Heat shock protocol and (D) reverse-transcription quantitative polymerase chain reaction (RT-qPCR) analysis of heat shock response genes in HCT116 cells following 1 h heat shock with/without OTS964 (156 nM).

(E) Average Pol II and CDK11 ChIP profiles for LPS response genes in THP1 cells treated with LPS (1 μg/mL) for 3 h and/or OTS964 (156 nM) for 2 h.

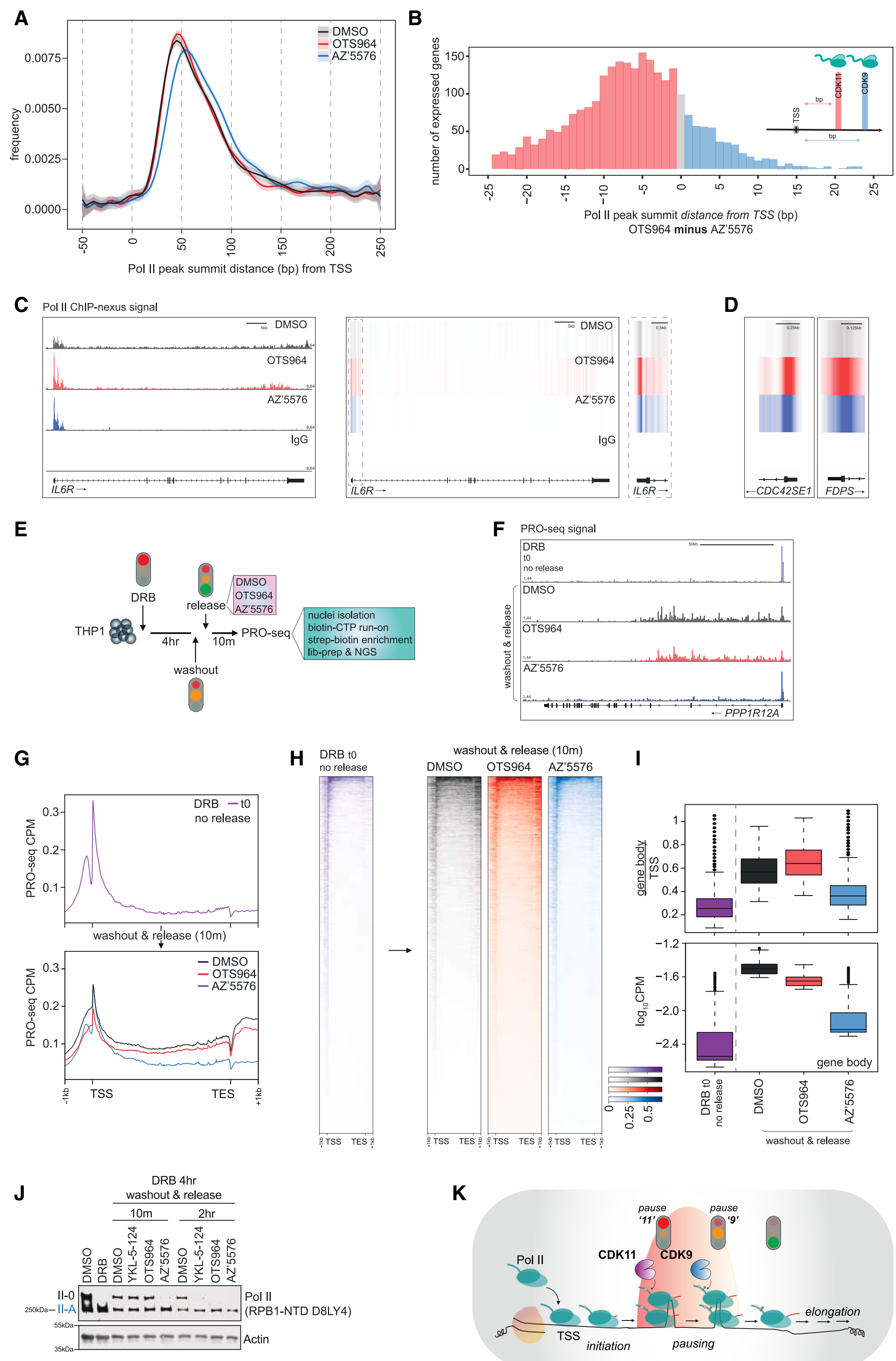
(F and G) Pol II ChIP (F) $\log_{10}$CPM coverage across TSS and gene body regions, and (G) ratio of $\log_{10}$CPM gene body vs. TSS coverage, for LPS response genes.

(H) Representative (*CXCL2* locus) Pol II and CDK11 ChIP-signal.

(B, E, F, and G) were analyzed by unpaired, two-sided Student's t test ***$p < 0.001$ and represent 460 LPS response genes (logFC > 1; adj. p value < 0.05).

(D) is representative of the mean ± SEM of five independent experiments and was analyzed by one-way ANOVA, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

the direct phosphorylation of the Pol II CTD by recombinant CDK11 confirms its biochemical capacity, further *in vivo* studies are needed to determine whether decreased Pol II and/or SPT5 phosphorylation following CDK11 inhibition is due to the direct absence of CDK11-deposited phospho-marks, consistent with a model of sequential CDK11- and CDK9-mediated phospho-regulation, or is instead a consequence of Pol II stalling prior to reaching the point where CDK9 phosphorylates these residues. Additionally, while it is also unknown whether the coordinated sequential activities of CDK11 and CDK9 would be necessary for the co-regulated expression of all genes, this study and previous reports have shown that selective inhibition of CDK7, CDK9, CDK11, and CDK12/13 have profound transcriptome-wide effects on gene expression and cell viability, suggesting that the different transcriptional CDKs cannot always compensate for each other. Our positioning of CDK11 within the broader



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transcriptional CDK network that governs Pol II transcription cycles is further supported by the observation that while CDK11 activity promotes transition of Pol II into gene bodies of LPS response genes, it is dispensable for the initial recruitment of Pol II to these loci. Additionally, in contrast to mNET-seq data demonstrating that CDK7 inhibition results in decreased coverage of transcriptionally engaged Pol II at TSS-proximal regions, reflective of a transcription initiation defect,[87] PRO-seq assays show an accumulation of transcriptionally engaged Pol II in response to CDK11 inhibition.

There is clear evidence that the Pol II pausing checkpoint should be best understood as a dynamic continuum, as opposed to a fixed variable or single stop-start point.[44–48] Previous studies employing base-pair resolution PRO-seq and eNET-seq approaches have described "early" and "late" Pol II pausing peaks separated by an estimated 10 nucleotides,[44,45] "first" and "second" pause sites with a median distance of 49 bp,[47] as well as high and low frequency pausing zones that can extend up to 3 kb into gene bodies.[48] Our results support a model of dynamic and sequential Pol II pause-release (Figure 7J), where CDK11 and CDK9 activities are proposed to be independently necessary for progression to transcription elongation, and inhibition of either kinase is sufficient to induce Pol II TSS accumulation and ablate nascent RNA synthesis. Intriguingly, the CDK11-dependent pause sites identified using high-resolution ChIP-nexus and PRO-seq approaches exhibit overlap, with respect to their distance from the TSS, with those in vehicle-treated cells. This raises the interesting possibility that, within a dynamic pausing regulatory zone, CDK11 may primarily regulate or at least contribute to the modulation of natural Pol II pausing, for which the exact molecular regulatory mechanism has yet to be fully elucidated.

Pol II accumulation observed following CDK11 inhibition may occur via a molecular feedback response to the disruption of spliceosome complex assembly and aberrant pre-mRNA splicing, as opposed to a more direct role for CDK11 for the control of Pol II near the TSS.[67] Links between disrupted pre-mRNA splicing and Pol II transcription are complex, with common cancer-associated *SF3B1* mutations resulting in early spliceosome assembly defects, and depletion of core spliceosome factors reported to reduce Pol II occupancy at TSS regions.[88,89] Catalytic CDK7 (Pol II initiation[90]), CDK9 (pausing[81]), and CDK12/13 (elongation[91,92]) inhibitions have also been reported to modify SF3B1 phosphorylation status and result in the aberrant expression of alternative transcript variants. Our findings support a model where CDK11 may operate to regulate Pol II transcription and gene expression, both through direct modulation of Pol II pause-release sites, as well as regulation of SF3B1-directed mRNA splicing across transcription cycles.

The rapid, robust impacts of OTS964 treatment in two mouse models of hematological cancer strongly suggest the importance of CDK11 for the viability of aggressive malignancies and support further evaluation of anti-CDK11 therapies using longer-term survival studies in pre-clinical human cancer models. While this work demonstrates the impact of CDK11 inhibition for malignancies characterized by genetic drivers associated with dysregulated transcription (*MLL*-rearranged AML, *MYC* amplified lymphoma), a requirement for CDK11 for survival and proliferation has been revealed across a broad range of solid and non-solid cancer cell types using RNAi and CRISPR-Cas9 genetic depletion approaches,[93–100] while *in vivo* efficacy of OTS964 has been shown in human lung cancer xenograft models.[66] Importantly, our findings highlight the importance of CDK11 transcriptional functions, as opposed to cell-cycle regulatory roles for the maintenance of cancer cell viability, with ectopic expression of the CDK11 cell-cycle variant insufficient to rescue AML cells following depletion of the transcriptional isoform.

Taken together, this study posits an important role for CDK11 in the transcription cycle, operating within a dynamic, multi-checkpoint pause-release zone, and precisely controlling steady-state and inducible gene expression. This work reveals an additional regulatory layer within the already complex network governing Pol II-dependent gene expression that may be strategically targeted for therapeutic benefit in aggressive cancers.

Limitations of the study

The *in vivo* experiments measured the impact of a single OTS964 dose on known biological indicators of *MLL*r-AML and Eμ-*Myc* B-lymphoma, including splenomegaly and increased WBC count, in addition to flow-cytometry analysis of fluorophore-labeled tumor cells. Further evaluation of extended OTS964 dosing in longer-term experiments will be necessary to determine if the observed short-term reductions in tumor burden result in extended survival and tangible therapeutic benefit. In xenograft models of human lung cancer, extended treatment of mice with OTS964 at a tolerated dose has been potentially associated with hematological toxicity (reduced WBC),[74] and future studies

Figure 7. CDK11-dependent Pol II pause site precedes CDK9-regulated pausing

(A and B) (A) Distance (bp) and (B) differences in the distance of Pol II ChIP-nexus peak summits from the TSS at individual genes in THP1 cells treated with DMSO, OTS964 (156 nM), and AZ′5576 (170 nM) for 2 h.

(C and D) Representative Pol II ChIP-nexus signal at (C) *IL6R*, (D) *CDC42SE1*, and *FDPS* loci.

(E and F) (E) DRB-release PRO-seq experiment and (F) representative PRO-seq signal (*PPP1R12A*) in THP1 cells treated with DRB (100 μM) for 4 h prior to washout/release with DMSO, OTS964 (156 nM), and AZ′5576 (170 nM) for 10 min.

(G and H) PRO-seq (G) average profiles and (H) metagene plots (n = 9,932 expressed genes).

(I) PRO-seq $\log_{10}$CPM gene body vs. TSS ratio and gene body coverage (n = 9,932 expressed genes).

(J) THP1 cells treated with DRB for 4 h prior to washout/release with DMSO, YKL-5-124 (125 nM), OTS964 (156 nM), and AZ′5576 (170 nM) for 10 min or 2 h. II-O and II-A represent the hyper- and hypo-phosphorylated Pol II variants.

(K) Proposed model.

(A–D) is representative of three replicate ChIPs.

(F–I) is representative of two independent PRO-seqs.

(J) is representative of three independent experiments.

should assess the impact of longer-term OTS964 treatment on non-tumor blood cells in aggressive blood cancer models.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents can be directed to the lead contact, Ricky W. Johnstone (ricky.johnstone@petermac.org).

Materials availability

This study did not generate any unique reagents.

Data and code availability

- Bioinformatics and proteomics datasets generated by this study are available from NCBI Gene Expression Omnibus (https://www.ncbi.nlm.nih.gov/geo/; GEO: GSE280102, GSE280103, and GSE280208) and ProteomeXchange Consortium/PRID (https://www.ebi.ac.uk/pride/; PRIDE: PXD066230).
- This paper does not report any original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

J.R.D. performed experiments, data analysis, and data interpretation. B.M., K.T., F.B., S.V., A.C., P.F., and A.M. performed experiments and data analysis. N.B., E.T., and A.H. performed bioinformatics analysis and data interpretation. J.R.D., Z.F., I.T., J.A.T., A.N., J.J.S., K.J.S., M.G., and S.J.V. provided critical reagents and protocols and contributed to experimental design and data interpretation. J.R.D. and R.W.J. wrote the manuscript, and R.W.J. supervised the study.

DECLARATION OF INTERESTS

The Johnstone lab receives research funding from Pfizer, BMS, MycRx, and AstraZeneca. R.W.J. is a co-founder and shareholder of MycRx and receives consultancy payments.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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REFERENCES

1. Parua, P.K., and Fisher, R.P. (2020). Dissecting the Pol II transcription cycle and derailing cancer with CDK inhibitors. Nat. Chem. Biol. *16*, 716–724. https://doi.org/10.1038/s41589-020-0563-4.
2. Vervoort, S.J., Devlin, J.R., Kwiatkowski, N., Teng, M., Gray, N.S., and Johnstone, R.W. (2022). Targeting transcription cycles in cancer. Nat. Rev. Cancer *22*, 5–24. https://doi.org/10.1038/s41568-021-00411-8.
3. Bradner, J.E., Hnisz, D., and Young, R.A. (2017). Transcriptional Addiction in Cancer. Cell *168*, 629–643. https://doi.org/10.1016/j.cell.2016.12.013.
4. Gonda, T.J., and Ramsay, R.G. (2015). Directly targeting transcriptional dysregulation in cancer. Nat. Rev. Cancer *15*, 686–694. https://doi.org/10.1038/nrc4018.
5. Okabe, A., and Kaneda, A. (2021). Transcriptional dysregulation by aberrant enhancer activation and rewiring in cancer. Cancer Sci. *112*, 2081–2088. https://doi.org/10.1111/cas.14884.
6. Hsin, J.P., and Manley, J.L. (2012). The RNA polymerase II CTD coordinates transcription and RNA processing. Genes Dev. *26*, 2119–2137. https://doi.org/10.1101/gad.200303.112.
7. Eick, D., and Geyer, M. (2013). The RNA polymerase II carboxy-terminal domain (CTD) code. Chem. Rev. *113*, 8456–8490. https://doi.org/10.1021/cr400071f.
8. Zaborowska, J., Egloff, S., and Murphy, S. (2016). The pol II CTD: new twists in the tail. Nat. Struct. Mol. Biol. *23*, 771–777. https://doi.org/10.1038/nsmb.3285.
9. Harlen, K.M., and Churchman, L.S. (2017). The code and beyond: transcription regulation by the RNA polymerase II carboxy-terminal domain. Nat. Rev. Mol. Cell Biol. *18*, 263–273. https://doi.org/10.1038/nrm.2017.10.

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10. Core, L.J., Waterfall, J.J., and Lis, J.T. (2008). Nascent RNA sequencing reveals widespread pausing and divergent initiation at human promoters. Science *322*, 1845–1848. https://doi.org/10.1126/science.1162228.

11. Core, L., and Adelman, K. (2019). Promoter-proximal pausing of RNA polymerase II: a nexus of gene regulation. Genes Dev. *33*, 960–982. https://doi.org/10.1101/gad.325142.119.

12. Abuhashem, A., Garg, V., and Hadjantonakis, A.K. (2022). RNA polymerase II pausing in development: orchestrating transcription. Open Biol. *12*, 210220. https://doi.org/10.1098/rsob.210220.

13. Dollinger, R., and Gilmour, D.S. (2021). Regulation of Promoter Proximal Pausing of RNA Polymerase II in Metazoans. J. Mol. Biol. *433*, 166897. https://doi.org/10.1016/j.jmb.2021.166897.

14. Aoi, Y., and Shilatifard, A. (2023). Transcriptional elongation control in developmental gene expression, aging, and disease. Mol. Cell *83*, 3972–3999. https://doi.org/10.1016/j.molcel.2023.10.004.

15. Chen, F.X., Smith, E.R., and Shilatifard, A. (2018). Born to run: control of transcription elongation by RNA polymerase II. Nat. Rev. Mol. Cell Biol. *19*, 464–478. https://doi.org/10.1038/s41580-018-0010-5.

16. Su, B.G., and Vos, S.M. (2024). Distinct negative elongation factor conformations regulate RNA polymerase II promoter-proximal pausing. Mol. Cell *84*, 1243–1256.e5. https://doi.org/10.1016/j.molcel.2024.01.023.

17. Zhan, Y., Grabbe, F., Oberbeckmann, E., Dienemann, C., and Cramer, P. (2024). Three-step mechanism of promoter escape by RNA polymerase II. Mol. Cell *84*, 1699–1710.e6. https://doi.org/10.1016/j.molcel.2024.03.016.

18. Wu, C.H., Yamaguchi, Y., Benjamin, L.R., Horvat-Gordon, M., Washinsky, J., Enerly, E., Larsson, J., Lambertsson, A., Handa, H., and Gilmour, D. (2003). NELF and DSIF cause promoter proximal pausing on the hsp70 promoter in Drosophila. Genes Dev. *17*, 1402–1414. https://doi.org/10.1101/gad.1091403.

19. Vos, S.M., Farnung, L., Urlaub, H., and Cramer, P. (2018). Structure of paused transcription complex Pol II–DSIF–NELF. Nature *560*, 601–606. https://doi.org/10.1038/s41586-018-0442-2.

20. Yamaguchi, Y., Takagi, T., Wada, T., Yano, K., Furuya, A., Sugimoto, S., Hasegawa, J., and Handa, H. (1999). NELF, a multisubunit complex containing RD, cooperates with DSIF to repress RNA polymerase II elongation. Cell *97*, 41–51. https://doi.org/10.1016/s0092-8674(00)80713-8.

21. Marshall, N.F., and Price, D.H. (1995). Purification of P-TEFb, a Transcription Factor Required for the Transition into Productive Elongation. J. Biol. Chem. *270*, 12335–12338. https://doi.org/10.1074/jbc.270.21.12335.

22. Wada, T., Takagi, T., Yamaguchi, Y., Watanabe, D., and Handa, H. (1998). Evidence that P-TEFb alleviates the negative effect of DSIF on RNA polymerase II-dependent transcription in vitro. EMBO J. *17*, 7395–7403. https://doi.org/10.1093/emboj/17.24.7395.

23. Kim, J.B., and Sharp, P.A. (2001). Positive Transcription Elongation Factor b Phosphorylates hSPT5 and RNA Polymerase II Carboxyl-terminal Domain Independently of Cyclin-dependent Kinase-activating Kinase. J. Biol. Chem. *276*, 12317–12323. https://doi.org/10.1074/jbc.M010908200.

24. Yamada, T., Yamaguchi, Y., Inukai, N., Okamoto, S., Mura, T., and Handa, H. (2006). P-TEFb-Mediated Phosphorylation of hSpt5 C-Terminal Repeats Is Critical for Processive Transcription Elongation. Mol. Cell *21*, 227–237. https://doi.org/10.1016/j.molcel.2005.11.024.

25. Booth, G.T., Parua, P.K., Sansó, M., Fisher, R.P., and Lis, J.T. (2018). Cdk9 regulates a promoter-proximal checkpoint to modulate RNA polymerase II elongation rate in fission yeast. Nat. Commun. *9*, 543. https://doi.org/10.1038/s41467-018-03006-4.

26. Rahl, P.B., Lin, C.Y., Seila, A.C., Flynn, R.A., McCuine, S., Burge, C.B., Sharp, P.A., and Young, R.A. (2010). c-Myc regulates transcriptional pause release. Cell *141*, 432–445. https://doi.org/10.1016/j.cell.2010.03.030.

27. Kanazawa, S., Soucek, L., Evan, G., Okamoto, T., and Peterlin, B.M. (2003). c-Myc recruits P-TEFb for transcription, cellular proliferation and apoptosis. Oncogene *22*, 5707–5711. https://doi.org/10.1038/sj.onc.1206800.

28. Lin, C.Y., Lovén, J., Rahl, P.B., Paranal, R.M., Burge, C.B., Bradner, J.E., Lee, T.I., and Young, R.A. (2012). Transcriptional amplification in tumor cells with elevated c-Myc. Cell *151*, 56–67. https://doi.org/10.1016/j.cell.2012.08.026.

29. Nie, Z., Hu, G., Wei, G., Cui, K., Yamane, A., Resch, W., Wang, R., Green, D.R., Tessarollo, L., Casellas, R., et al. (2012). c-Myc is a universal amplifier of expressed genes in lymphocytes and embryonic stem cells. Cell *151*, 68–79. https://doi.org/10.1016/j.cell.2012.08.033.

30. Baker, A., Gregory, G.P., Verbrugge, I., Kats, L., Hilton, J.J., Vidacs, E., Lee, E.M., Lock, R.B., Zuber, J., Shortt, J., et al. (2016). The CDK9 Inhibitor Dinaciclib Exerts Potent Apoptotic and Antitumor Effects in Preclinical Models of MLL-Rearranged Acute Myeloid Leukemia. Cancer Res. *76*, 1158–1169. https://doi.org/10.1158/0008-5472.CAN-15-1070.

31. Gregory, G.P., Hogg, S.J., Kats, L.M., Vidacs, E., Baker, A.J., Gilan, O., Lefebure, M., Martin, B.P., Dawson, M.A., Johnstone, R.W., et al. (2015). CDK9 inhibition by dinaciclib potently suppresses Mcl-1 to induce durable apoptotic responses in aggressive MYC-driven B-cell lymphoma in vivo. Leukemia *29*, 1437–1441. https://doi.org/10.1038/leu.2015.10.

32. Fong, C.Y., Gilan, O., Lam, E.Y.N., Rubin, A.F., Ftouni, S., Tyler, D., Stanley, K., Sinha, D., Yeh, P., Morison, J., et al. (2015). BET inhibitor resistance emerges from leukaemia stem cells. Nature *525*, 538–542. https://doi.org/10.1038/nature14888.

33. Dawson, M.A., Prinjha, R.K., Dittmann, A., Giotopoulos, G., Bantscheff, M., Chan, W.I., Robson, S.C., Chung, C.W., Hopf, C., Savitski, M.M., et al. (2011). Inhibition of BET recruitment to chromatin as an effective treatment for MLL-fusion leukaemia. Nature *478*, 529–533. https://doi.org/10.1038/nature10509.

34. Hogg, S.J., Newbold, A., Vervoort, S.J., Cluse, L.A., Martin, B.P., Gregory, G.P., Lefebure, M., Vidacs, E., Tothill, R.W., Bradner, J.E., et al. (2016). BET Inhibition Induces Apoptosis in Aggressive B-Cell Lymphoma via Epigenetic Regulation of BCL-2 Family Members. Mol. Cancer Ther. *15*, 2030–2041. https://doi.org/10.1158/1535-7163.MCT-15-0924.

35. Vos, S.M., Farnung, L., Boehning, M., Wigge, C., Linden, A., Urlaub, H., and Cramer, P. (2018). Structure of activated transcription complex Pol II–DSIF–PAF–SPT6. Nature *560*, 607–612. https://doi.org/10.1038/s41586-018-0440-4.

36. Fianu, I., Chen, Y., Dienemann, C., Dybkov, O., Linden, A., Urlaub, H., and Cramer, P. (2021). Structural basis of Integrator-mediated transcription regulation. Science *374*, 883–887. https://doi.org/10.1126/science.abk0154.

37. Parua, P.K., Kalan, S., Benjamin, B., Sansó, M., and Fisher, R.P. (2020). Distinct Cdk9-phosphatase switches act at the beginning and end of elongation by RNA polymerase II. Nat. Commun. *11*, 4338. https://doi.org/10.1038/s41467-020-18173-6.

38. Cossa, G., Parua, P.K., Eilers, M., and Fisher, R.P. (2021). Protein phosphatases in the RNAPII transcription cycle: erasers, sculptors, gatekeepers, and potential drug targets. Genes Dev. *35*, 658–676. https://doi.org/10.1101/gad.348315.121.

39. Zheng, H., Qi, Y., Hu, S., Cao, X., Xu, C., Yin, Z., Chen, X., Li, Y., Liu, W., Li, J., et al. (2020). Identification of Integrator-PP2A complex (INTAC), an RNA polymerase II phosphatase. Science *370*, eabb5872. https://doi.org/10.1126/science.abb5872.

40. Hu, S., Peng, L., Song, A., Ji, Y.X., Cheng, J., Wang, M., and Chen, F.X. (2023). INTAC endonuclease and phosphatase modules differentially regulate transcription by RNA polymerase II. Mol. Cell *83*, 1588–1604.e5. https://doi.org/10.1016/j.molcel.2023.03.022.

41. Fujiwara, R., Zhai, S.N., Liang, D., Shah, A.P., Tracey, M., Ma, X.K., Fields, C.J., Mendoza-Figueroa, M.S., Meline, M.C., Tatomer, D.C., et al. (2023). IntS6 and the Integrator phosphatase module tune the efficiency of select premature transcription termination events. Mol. Cell *83*, 4445–4460.e7. https://doi.org/10.1016/j.molcel.2023.10.035.

42. Beckedorff, F., Blumenthal, E., daSilva, L.F., Aoi, Y., Cingaram, P.R., Yue, J., Zhang, A., Dokaneheifard, S., Valencia, M.G., Gaidosh, G., et al. (2020). The Human Integrator Complex Facilitates Transcriptional Elongation by Endonucleolytic Cleavage of Nascent Transcripts. Cell Rep. *32*, 107917. https://doi.org/10.1016/j.celrep.2020.107917.

43. Lykke-Andersen, S., Žumer, K., Molska, E.Š., Rouvière, J.O., Wu, G., Demel, C., Schwalb, B., Schmid, M., Cramer, P., and Jensen, T.H. (2021). Integrator is a genome-wide attenuator of non-productive transcription. Mol. Cell *81*, 514–529.e6. https://doi.org/10.1016/j.molcel.2020.12.014.

44. Vihervaara, A., Versluis, P., Himanen, S.V., and Lis, J.T. (2023). PRO-IP-seq tracks molecular modifications of engaged Pol II complexes at nucleotide resolution. Nat. Commun. *14*, 7039. https://doi.org/10.1038/s41467-023-42715-3.

45. Rasmussen, E.B., and Lis, J.T. (1993). In vivo transcriptional pausing and cap formation on three Drosophila heat shock genes. Proc. Natl. Acad. Sci. USA *90*, 7923–7927. https://doi.org/10.1073/pnas.90.17.7923.

46. Tome, J.M., Tippens, N.D., and Lis, J.T. (2018). Single-molecule nascent RNA sequencing identifies regulatory domain architecture at promoters and enhancers. Nat. Genet. *50*, 1533–1541. https://doi.org/10.1038/s41588-018-0234-5.

47. Aoi, Y., Smith, E.R., Shah, A.P., Rendleman, E.J., Marshall, S.A., Woodfin, A.R., Chen, F.X., Shiekhattar, R., and Shilatifard, A. (2020). NELF Regulates a Promoter-Proximal Step Distinct from RNA Pol II Pause-Release. Mol. Cell *78*, 261–274.e5. https://doi.org/10.1016/j.molcel.2020.02.014.

48. Fong, N., Sheridan, R.M., Ramachandran, S., and Bentley, D.L. (2022). The pausing zone and control of RNA polymerase II elongation by Spt5: Implications for the pause-release model. Mol. Cell *82*, 3632–3645.e4. https://doi.org/10.1016/j.molcel.2022.09.001.

49. DeBerardine, M., Booth, G.T., Versluis, P.P., and Lis, J.T. (2023). The NELF pausing checkpoint mediates the functional divergence of Cdk9. Nat. Commun. *14*, 2762. https://doi.org/10.1038/s41467-023-38359-y.

50. Aoi, Y., Takahashi, Y.H., Shah, A.P., Iwanaszko, M., Rendleman, E.J., Khan, N.H., Cho, B.K., Goo, Y.A., Ganesan, S., Kelleher, N.L., et al. (2021). SPT5 stabilization of promoter-proximal RNA polymerase II. Mol. Cell *81*, 4413–4424.e5. https://doi.org/10.1016/j.molcel.2021.08.006.

51. Hu, S., Peng, L., Xu, C., Wang, Z., Song, A., and Chen, F.X. (2021). SPT5 stabilizes RNA polymerase II, orchestrates transcription cycles, and maintains the enhancer landscape. Mol. Cell *81*, 4425–4439.e6. https://doi.org/10.1016/j.molcel.2021.08.029.

52. Wang, Z., Song, A., Xu, H., Hu, S., Tao, B., Peng, L., Wang, J., Li, J., Yu, J., Wang, L., et al. (2022). Coordinated regulation of RNA polymerase II pausing and elongation progression by PAF1. Sci. Adv. *8*, eabm5504. https://doi.org/10.1126/sciadv.abm5504.

53. Xu, Y., Bernecky, C., Lee, C.T., Maier, K.C., Schwalb, B., Tegunov, D., Plitzko, J.M., Urlaub, H., and Cramer, P. (2017). Architecture of the RNA polymerase II-Paf1C-TFIIS transcription elongation complex. Nat. Commun. *8*, 15741. https://doi.org/10.1038/ncomms15741.

54. Chen, F.X., Xie, P., Collings, C.K., Cao, K., Aoi, Y., Marshall, S.A., Rendleman, E.J., Ugarenko, M., Ozark, P.A., Zhang, A., et al. (2017). PAF1 regulation of promoter-proximal pause release via enhancer activation. Science *357*, 1294–1298. https://doi.org/10.1126/science.aan3269.

55. Chen, F.X., Woodfin, A.R., Gardini, A., Rickels, R.A., Marshall, S.A., Smith, E.R., Shiekhattar, R., and Shilatifard, A. (2015). PAF1, a Molecular Regulator of Promoter-Proximal Pausing by RNA Polymerase II. Cell *162*, 1003–1015. https://doi.org/10.1016/j.cell.2015.07.042.

56. Yu, M., Yang, W., Ni, T., Tang, Z., Nakadai, T., Zhu, J., and Roeder, R.G. (2015). RNA polymerase II-associated factor 1 regulates the release and phosphorylation of paused RNA polymerase II. Science *350*, 1383–1386. https://doi.org/10.1126/science.aad2338.

57. Žumer, K., Maier, K.C., Farnung, L., Jaeger, M.G., Rus, P., Winter, G., and Cramer, P. (2021). Two distinct mechanisms of RNA polymerase II elongation stimulation in vivo. Mol. Cell *81*, 3096–3109.e8. https://doi.org/10.1016/j.molcel.2021.05.028.

58. Žumer, K., Ochmann, M., Aljahani, A., Zheenbekova, A., Devadas, A., Maier, K.C., Rus, P., Neef, U., Oudelaar, A.M., and Cramer, P. (2024). FACT maintains chromatin architecture and thereby stimulates RNA polymerase II pausing during transcription in vivo. Mol. Cell *84*, 2053–2069.e9. https://doi.org/10.1016/j.molcel.2024.05.003.

59. Aoi, Y., Shah, A.P., Ganesan, S., Soliman, S.H.A., Cho, B.-K., Goo, Y.A., Kelleher, N.L., and Shilatifard, A. (2022). SPT6 functions in transcriptional pause/release via PAF1C recruitment. Mol. Cell *82*, 3412–3423.e5. https://doi.org/10.1016/j.molcel.2022.06.037.

60. Tettey, T.T., Gao, X., Shao, W., Li, H., Story, B.A., Chitsazan, A.D., Glaser, R.L., Goode, Z.H., Seidel, C.W., Conaway, R.C., et al. (2019). A Role for FACT in RNA Polymerase II Promoter-Proximal Pausing. Cell Rep. *27*, 3770–3779.e7. https://doi.org/10.1016/j.celrep.2019.05.099.

61. Singh, S., Szlachta, K., Manukyan, A., Raimer, H.M., Dinda, M., Bekiranov, S., and Wang, Y.H. (2020). Pausing sites of RNA polymerase II on actively transcribed genes are enriched in DNA double-stranded breaks. J. Biol. Chem. *295*, 3990–4000. https://doi.org/10.1074/jbc.RA119.011665.

62. Herrero-Ruiz, A., Martínez-García, P.M., Terrón-Bautista, J., Millán-Zambrano, G., Lieberman, J.A., Jimeno-González, S., and Cortés-Ledesma, F. (2021). Topoisomerase IIα represses transcription by enforcing promoter-proximal pausing. Cell Rep. *35*, 108977. https://doi.org/10.1016/j.celrep.2021.108977.

63. Bunch, H., Kim, D., Naganuma, M., Nakagawa, R., Cong, A., Jeong, J., Ehara, H., Vu, H., Chang, J.H., Schellenberg, M.J., et al. (2023). ERK2-topoisomerase II regulatory axis is important for gene activation in immediate early genes. Nat. Commun. *14*, 8341. https://doi.org/10.1038/s41467-023-44089-y.

64. Zhou, Y., Shen, J.K., Hornicek, F.J., Kan, Q., and Duan, Z. (2016). The emerging roles and therapeutic potential of cyclin-dependent kinase 11 (CDK11) in human cancer. Oncotarget *7*, 40846–40859. https://doi.org/10.18632/oncotarget.8519.

65. Kelso, S., O'Brien, S., Kurinov, I., Angers, S., and Sicheri, F. (2022). Crystal structure of the CDK11 kinase domain bound to the small-molecule inhibitor OTS964. Structure *30*, 1615–1625.e4. https://doi.org/10.1016/j.str.2022.10.003.

66. Lin, A., Giuliano, C.J., Palladino, A., John, K.M., Abramowicz, C., Yuan, M.L., Sausville, E.L., Lukow, D.A., Liu, L., Chait, A.R., et al. (2019). Off-target toxicity is a common mechanism of action of cancer drugs undergoing clinical trials. Sci. Transl. Med. *11*, eaaw8412. https://doi.org/10.1126/scitranslmed.aaw8412.

67. Hluchý, M., Gajdušková, P., Ruiz de los Mozos, I., Rájecký, M., Kluge, M., Berger, B.-T., Slabá, Z., Potěšil, D., Weiß, E., Ule, J., et al. (2022). CDK11 regulates pre-mRNA splicing by phosphorylation of SF3B1. Nature *609*, 829–834. https://doi.org/10.1038/s41586-022-05204-z.

68. Wang, C., Xu, L., Du, C., Yun, H., Wang, K., Liu, H., Ye, M., Fan, J., Zhou, Y., and Cheng, H. (2023). CDK11 requires a critical activator SAP30BP to regulate pre-mRNA splicing. EMBO J. *42*, e114051. https://doi.org/10.15252/embj.2023114051.

69. Hu, D., Mayeda, A., Trembley, J.H., Lahti, J.M., and Kidd, V.J. (2003). CDK11 Complexes Promote Pre-mRNA Splicing. J. Biol. Chem. *278*, 8623–8629. https://doi.org/10.1074/jbc.M210057200.

70. Gajdušková, P., Ruiz de los Mozos, I., Rájecký, M., Hluchý, M., Ule, J., and Blazek, D. (2020). CDK11 is required for transcription of replication-dependent histone genes. Nat. Struct. Mol. Biol. *27*, 500–510. https://doi.org/10.1038/s41594-020-0406-8.

71. Zhang, Q., Chen, Y., Teng, Z., Lin, Z., and Liu, H. (2024). CDK11 facilitates centromeric transcription to maintain centromeric cohesion during mitosis. Mol. Biol. Cell *35*, ar18. https://doi.org/10.1091/mbc.E23-08-0303.

72. Feng, Y., Liao, Y., Zhang, J., Shen, J., Shao, Z., Hornicek, F., and Duan, Z. (2019). Transcriptional activation of CBFβ by CDK11p110 is necessary

to promote osteosarcoma cell proliferation. Cell Commun. Signal. *17*, 125. https://doi.org/10.1186/s12964-019-0440-5.

73. Pak, V., Eifler, T.T., Jäger, S., Krogan, N.J., Fujinaga, K., and Peterlin, B.M. (2015). CDK11 in TREX/THOC Regulates HIV mRNA 3′ End Processing. Cell Host Microbe *18*, 560–570. https://doi.org/10.1016/j.chom.2015.10.012.

74. Matsuo, Y., Park, J.H., Miyamoto, T., Yamamoto, S., Hisada, S., Alachkar, H., and Nakamura, Y. (2014). TOPK inhibitor induces complete tumor regression in xenograft models of human cancer through inhibition of cytokinesis. Sci. Transl. Med. *6*, 259ra145. https://doi.org/10.1126/scitranslmed.3010277.

75. DepMap, B. (2020) DepMap 20Q4 Public. Figshare.

76. Meyers, R.M., Bryan, J.G., McFarland, J.M., Weir, B.A., Sizemore, A.E., Xu, H., Dharia, N.V., Montgomery, P.G., Cowley, G.S., Pantel, S., et al. (2017). Computational correction of copy number effect improves specificity of CRISPR–Cas9 essentiality screens in cancer cells. Nat. Genet. *49*, 1779–1784. https://doi.org/10.1038/ng.3984.

77. Tsherniak, A., Vazquez, F., Montgomery, P.G., Weir, B.A., Kryukov, G., Cowley, G.S., Gill, S., Harrington, W.F., Pantel, S., Krill-Burger, J.M., et al. (2017). Defining a Cancer Dependency Map. Cell *170*, 564–576.e16. https://doi.org/10.1016/j.cell.2017.06.010.

78. Adams, J.M., Harris, A.W., Pinkert, C.A., Corcoran, L.M., Alexander, W.S., Cory, S., Palmiter, R.D., and Brinster, R.L. (1985). The c-myc oncogene driven by immunoglobulin enhancers induces lymphoid malignancy in transgenic mice. Nature *318*, 533–538. https://doi.org/10.1038/318533a0.

79. Herzog, V.A., Reichholf, B., Neumann, T., Rescheneder, P., Bhat, P., Burkard, T.R., Wlotzka, W., von Haeseler, A., Zuber, J., and Ameres, S. L. (2017). Thiol-linked alkylation of RNA to assess expression dynamics. Nat. Methods *14*, 1198–1204. https://doi.org/10.1038/nmeth.4435.

80. Schwalb, B., Michel, M., Zacher, B., Frühauf, K., Demel, C., Tresch, A., Gagneur, J., and Cramer, P. (2016). TT-seq maps the human transient transcriptome. Science *352*, 1225–1228. https://doi.org/10.1126/science.aad9841.

81. Vervoort, S.J., Welsh, S.A., Devlin, J.R., Barbieri, E., Knight, D.A., Offley, S., Bjelosevic, S., Costacurta, M., Todorovski, I., Kearney, C.J., et al. (2021). The PP2A-Integrator-CDK9 axis fine-tunes transcription and can be targeted therapeutically in cancer. Cell *184*, 3143–3162.e32. https://doi.org/10.1016/j.cell.2021.04.022.

82. Mahat, D.B., Kwak, H., Booth, G.T., Jonkers, I.H., Danko, C.G., Patel, R.K., Waters, C.T., Munson, K., Core, L.J., and Lis, J.T. (2016). Base-pair-resolution genome-wide mapping of active RNA polymerases using precision nuclear run-on (PRO-seq). Nat. Protoc. *11*, 1455–1476. https://doi.org/10.1038/nprot.2016.086.

83. Fukumura, K., Sperotto, L., Seuß, S., Kang, H.S., Yoshimoto, R., Sattler, M., and Mayeda, A. (2023). SAP30BP interacts with RBM17/SPF45 to promote splicing in a subset of human short introns. Cell Rep. *42*, 113534. https://doi.org/10.1016/j.celrep.2023.113534.

84. He, Q., Johnston, J., and Zeitlinger, J. (2015). ChIP-nexus enables improved detection of in vivo transcription factor binding footprints. Nat. Biotechnol. *33*, 395–401. https://doi.org/10.1038/nbt.3121.

85. Blears, D., Lou, J., Fong, N., Mitter, R., Sheridan, R.M., He, D., Dirac-Svejstrup, A.B., Bentley, D., and Svejstrup, J.Q. (2024). Redundant pathways for removal of defective RNA polymerase II complexes at a promoter-proximal pause checkpoint. Mol. Cell *84*, 4790–4807.e11. https://doi.org/10.1016/j.molcel.2024.10.012.

86. Cacioppo, R., Gillis, A., Shlamovitz, I., Zeller, A., Castiblanco, D., Crisp, A., Haworth, B., Arabiotorre, A., Abyaneh, P., Bao, Y., et al. (2024). CRL3ARMC5 ubiquitin ligase and Integrator phosphatase form parallel mechanisms to control early stages of RNA Pol II transcription. Mol. Cell *84*, 4808–4823.e13. https://doi.org/10.1016/j.molcel.2024.11.024.

87. Velychko, T., Mohammad, E., Ferrer-Vicens, I., Parfentev, I., Werner, M., Studniarek, C., Schwalb, B., Urlaub, H., Murphy, S., Cramer, P., et al. (2024). CDK7 kinase activity promotes RNA polymerase II promoter escape by facilitating initiation factor release. Mol. Cell *84*, 2287–2303.e10. https://doi.org/10.1016/j.molcel.2024.05.007.

88. Boddu, P.C., Gupta, A.K., Roy, R., De La Peña Avalos, B., Olazabal-Herrero, A., Neuenkirchen, N., Zimmer, J.T., Chandhok, N.S., King, D., Nannya, Y., et al. (2024). Transcription elongation defects link oncogenic SF3B1 mutations to targetable alterations in chromatin landscape. Mol. Cell *84*, 1475–1495.e18. https://doi.org/10.1016/j.molcel.2024.02.032.

89. Du, X., Qin, W., Yang, C., Dai, L., San, M., Xia, Y., Zhou, S., Wang, M., Wu, S., Zhang, S., et al. (2024). RBM22 regulates RNA polymerase II 5′ pausing, elongation rate, and termination by coordinating 7SK-P-TEFb complex and SPT5. Genome Biol. *25*, 102. https://doi.org/10.1186/s13059-024-03242-6.

90. Rimel, J.K., Poss, Z.C., Erickson, B., Maas, Z.L., Ebmeier, C.C., Johnson, J.L., Decker, T.M., Yaron, T.M., Bradley, M.J., Hamman, K.B., et al. (2020). Selective inhibition of CDK7 reveals high-confidence targets and new models for TFIIH function in transcription. Genes Dev. *34*, 1452–1473. https://doi.org/10.1101/gad.341545.120.

91. Fan, Z., Devlin, J.R., Hogg, S.J., Doyle, M.A., Harrison, P.F., Todorovski, I., Cluse, L.A., Knight, D.A., Sandow, J.J., Gregory, G., et al. (2020). CDK13 cooperates with CDK12 to control global RNA polymerase II processivity. Sci. Adv. *6*, eaaz5041. https://doi.org/10.1126/sciadv.aaz5041.

92. Krajewska, M., Dries, R., Grassetti, A.V., Dust, S., Gao, Y., Huang, H., Sharma, B., Day, D.S., Kwiatkowski, N., Pomaville, M., et al. (2019). CDK12 loss in cancer cells affects DNA damage response genes through premature cleavage and polyadenylation. Nat. Commun. *10*, 1757. https://doi.org/10.1038/s41467-019-09703-y.

93. Duan, Z., Zhang, J., Choy, E., Harmon, D., Liu, X., Nielsen, P., Mankin, H., Gray, N.S., and Hornicek, F.J. (2012). Systematic kinome shRNA screening identifies CDK11 (PITSLRE) kinase expression is critical for osteosarcoma cell growth and proliferation. Clin. Cancer Res. *18*, 4580–4588. https://doi.org/10.1158/1078-0432.CCR-12-1157.

94. Jia, B., Choy, E., Cote, G., Harmon, D., Ye, S., Kan, Q., Mankin, H., Hornicek, F., and Duan, Z. (2014). Cyclin-dependent kinase 11 (CDK11) is crucial in the growth of liposarcoma cells. Cancer Lett. *342*, 104–112. https://doi.org/10.1016/j.canlet.2013.08.040.

95. Zhou, Y., Han, C., Li, D., Yu, Z., Li, F., Li, F., An, Q., Bai, H., Zhang, X., Duan, Z., et al. (2015). Cyclin-dependent kinase 11(p110) (CDK11 (p110)) is crucial for human breast cancer cell proliferation and growth. Sci. Rep. *5*, 10433. https://doi.org/10.1038/srep10433.

96. Tiedemann, R.E., Zhu, Y.X., Schmidt, J., Yin, H., Shi, C.X., Que, Q., Basu, G., Azorsa, D., Perkins, L.M., Braggio, E., et al. (2010). Kinome-wide RNAi studies in human multiple myeloma identify vulnerable kinase targets, including a lymphoid-restricted kinase, GRK6. Blood *115*, 1594–1604. https://doi.org/10.1182/blood-2009-09-243980.

97. Kren, B.T., Unger, G.M., Abedin, M.J., Vogel, R.I., Henzler, C.M., Ahmed, K., and Trembley, J.H. (2015). Preclinical evaluation of cyclin dependent kinase 11 and casein kinase 2 survival kinases as RNA interference targets for triple negative breast cancer therapy. Breast Cancer Res. *17*, 19. https://doi.org/10.1186/s13058-015-0524-0.

98. Liu, X., Gao, Y., Shen, J., Yang, W., Choy, E., Mankin, H., Hornicek, F.J., and Duan, Z. (2016). Cyclin-Dependent Kinase 11 (CDK11) Is Required for Ovarian Cancer Cell Growth In Vitro and In Vivo, and Its Inhibition Causes Apoptosis and Sensitizes Cells to Paclitaxel. Mol. Cancer Ther. *15*, 1691–1701. https://doi.org/10.1158/1535-7163.MCT-16-0032.

99. Ahmed, R.L., Shaughnessy, D.P., Knutson, T.P., Vogel, R.I., Ahmed, K., Kren, B.T., and Trembley, J.H. (2019). CDK11 Loss Induces Cell Cycle Dysfunction and Death of BRAF and NRAS Melanoma Cells. Pharmaceuticals (Basel) *12*, 50. https://doi.org/10.3390/ph12020050.

100. Du, Y., Yan, D., Yuan, Y., Xu, J., Wang, S., Yang, Z., Cheng, W., Tian, X., and Kan, Q. (2019). CDK11(p110) plays a critical role in the tumorigenicity of esophageal squamous cell carcinoma cells and is a potential drug target. Cell Cycle *18*, 452–466. https://doi.org/10.1080/15384101.2019.1577665.

101. Sansó, M., Levin, R.S., Lipp, J.J., Wang, V.Y.F., Greifenberg, A.K., Quezada, E.M., Ali, A., Ghosh, A., Larochelle, S., Rana, T.M., et al. (2016). P-TEFb regulation of transcription termination factor Xrn2 revealed by a chemical genetic screen for Cdk9 substrates. Genes Dev. *30*, 117–131. https://doi.org/10.1101/gad.269589.115.

102. Hashiguchi, T., Bruss, N., Best, S., Lam, V., Danilova, O., Paiva, C.J., Wolf, J., Gilbert, E.W., Okada, C.Y., Kaur, P., et al. (2019). Cyclin-Dependent Kinase-9 Is a Therapeutic Target in MYC-Expressing Diffuse Large B-Cell Lymphoma. Mol. Cancer Ther. *18*, 1520–1532. https://doi.org/10.1158/1535-7163.MCT-18-1023.

103. Shortt, J., Martin, B.P., Newbold, A., Hannan, K.M., Devlin, J.R., Baker, A.J., Ralli, R., Cullinane, C., Schmitt, C.A., Reimann, M., et al. (2013). Combined inhibition of PI3K-related DNA damage response kinases and mTORC1 induces apoptosis in MYC-driven B-cell lymphomas. Blood *121*, 2964–2974. https://doi.org/10.1182/blood-2012-08-446096.

104. Doench, J.G., Fusi, N., Sullender, M., Hegde, M., Vaimberg, E.W., Donovan, K.F., Smith, I., Tothova, Z., Wilen, C., Orchard, R., et al. (2016). Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. Nat. Biotechnol. *34*, 184–191. https://doi.org/10.1038/nbt.3437.

105. Aubrey, B.J., Kelly, G.L., Kueh, A.J., Brennan, M.S., O'Connor, L., Milla, L., Wilcox, S., Tai, L., Strasser, A., and Herold, M.J. (2015). An inducible lentiviral guide RNA platform enables the identification of tumor-essential genes and tumor-promoting mutations in vivo. Cell Rep. *10*, 1422–1432. https://doi.org/10.1016/j.celrep.2015.02.002.

106. Reya, T., Duncan, A.W., Ailles, L., Domen, J., Scherer, D.C., Willert, K., Hintz, L., Nusse, R., and Weissman, I.L. (2003). A role for Wnt signalling in self-renewal of haematopoietic stem cells. Nature *423*, 409–414. https://doi.org/10.1038/nature01593.

107. Dull, T., Zufferey, R., Kelly, M., Mandel, R.J., Nguyen, M., Trono, D., and Naldini, L. (1998). A third-generation lentivirus vector with a conditional packaging system. J. Virol. *72*, 8463–8471. https://doi.org/10.1128/JVI.72.11.8463-8471.1998.

108. Lambrus, B.G., Moyer, T.C., and Holland, A.J. (2018). Applying the auxin-inducible degradation system for rapid protein depletion in mammalian cells. Methods Cell Biol. *144*, 107–135. https://doi.org/10.1016/bs.mcb.2018.03.004.

109. ORFeome Collaboration (2016). The ORFeome Collaboration: a genome-scale human ORF-clone resource. Nat. Methods *13*, 191–192. https://doi.org/10.1038/nmeth.3776.

110. Natsume, T., Kiyomitsu, T., Saga, Y., and Kanemaki, M.T. (2016). Rapid Protein Depletion in Human Cells by Auxin-Inducible Degron Tagging with Short Homology Donors. Cell Rep. *15*, 210–218. https://doi.org/10.1016/j.celrep.2016.03.001.

111. Rosenbluh, J., Mercer, J., Shrestha, Y., Oliver, R., Tamayo, P., Doench, J. G., Tirosh, I., Piccioni, F., Hartenian, E., Horn, H., et al. (2016). Genetic and Proteomic Interrogation of Lower Confidence Candidate Genes Reveals Signaling Networks in β-Catenin-Active Cancers. Cell Syst. *3*, 302–316.e4. https://doi.org/10.1016/j.cels.2016.09.001.

112. Wang, Y., Dow, E.C., Liang, Y.Y., Ramakrishnan, R., Liu, H., Sung, T.L., Lin, X., and Rice, A.P. (2008). Phosphatase PPM1A regulates phosphorylation of Thr-186 in the Cdk9 T-loop. J. Biol. Chem. *283*, 33578–33584. https://doi.org/10.1074/jbc.M807495200.

113. Sakaue-Sawano, A., Kurokawa, H., Morimura, T., Hanyu, A., Hama, H., Osawa, H., Kashiwagi, S., Fukami, K., Miyata, T., Miyoshi, H., et al. (2008). Visualizing Spatiotemporal Dynamics of Multicellular Cell-Cycle Progression. Cell *132*, 487–498. https://doi.org/10.1016/j.cell.2007.12.033.

114. Kim, D., Langmead, B., and Salzberg, S.L. (2015). HISAT: a fast spliced aligner with low memory requirements. Nat. Methods *12*, 357–360. https://doi.org/10.1038/nmeth.3317.

115. Liao, Y., Smyth, G.K., and Shi, W. (2013). The Subread aligner: fast, accurate and scalable read mapping by seed-and-vote. Nucleic Acids Res. *41*, e108. https://doi.org/10.1093/nar/gkt214.

116. Langmead, B., and Salzberg, S.L. (2012). Fast gapped-read alignment with Bowtie 2. Nat. Methods *9*, 357–359. https://doi.org/10.1038/nmeth.1923.

117. Li, H., Handsaker, B., Wysoker, A., Fennell, T., Ruan, J., Homer, N., Marth, G., Abecasis, G., and Durbin, R.; 1000 Genome Project Data Processing Subgroup (2009). The Sequence Alignment/Map format and SAMtools. Bioinformatics *25*, 2078–2079. https://doi.org/10.1093/bioinformatics/btp352.

118. Ramírez, F., Dündar, F., Diehl, S., Grüning, B.A., and Manke, T. (2014). deepTools: a flexible platform for exploring deep-sequencing data. Nucleic Acids Res. *42*, W187–W191. https://doi.org/10.1093/nar/gku365.

119. Zhang, Y., Liu, T., Meyer, C.A., Eeckhoute, J., Johnson, D.S., Bernstein, B.E., Nusbaum, C., Myers, R.M., Brown, M., Li, W., et al. (2008). Model-based Analysis of ChIP-Seq (MACS). Genome Biol. *9*, R137. https://doi.org/10.1186/gb-2008-9-9-r137.

120. Dobin, A., Davis, C.A., Schlesinger, F., Drenkow, J., Zaleski, C., Jha, S., Batut, P., Chaisson, M., and Gingeras, T.R. (2013). STAR: ultrafast universal RNA-seq aligner. Bioinformatics *29*, 15–21. https://doi.org/10.1093/bioinformatics/bts635.

121. Neumann, T., Herzog, V.A., Muhar, M., von Haeseler, A., Zuber, J., Ameres, S.L., and Rescheneder, P. (2019). Quantification of experimentally induced nucleotide conversions in high-throughput sequencing datasets. BMC Bioinformatics *20*, 258. https://doi.org/10.1186/s12859-019-2849-7.

122. Bywater, M.J., Poortinga, G., Sanij, E., Hein, N., Peck, A., Cullinane, C., Wall, M., Cluse, L., Drygin, D., Anderes, K., et al. (2012). Inhibition of RNA polymerase I as a therapeutic strategy to promote cancer-specific activation of p53. Cancer Cell *22*, 51–65. https://doi.org/10.1016/j.ccr.2012.05.019.

123. Cox, J., and Mann, M. (2008). MaxQuant enables high peptide identification rates, individualized p.p.b.-range mass accuracies and proteome-wide protein quantification. Nat. Biotechnol. *26*, 1367–1372. https://doi.org/10.1038/nbt.1511.

124. Kuhn, L., Vincent, T., Hammann, P., and Zuber, H. (2023). Exploring Protein Interactome Data with IPinquiry: Statistical Analysis and Data Visualization by Spectral Counts. Methods Mol. Biol. *2426*, 243–265. https://doi.org/10.1007/978-1-0716-1967-4_11.

125. Szklarczyk, D., Kirsch, R., Koutrouli, M., Nastou, K., Mehryary, F., Hachilif, R., Gable, A.L., Fang, T., Doncheva, N.T., Pyysalo, S., et al. (2023). The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. Nucleic Acids Res. *51*, D638–D646. https://doi.org/10.1093/nar/gkac1000.

126. Perez-Riverol, Y., Bandla, C., Kundu, D.J., Kamatchinathan, S., Bai, J., Hewapathirana, S., John, N.S., Prakash, A., Walzer, M., Wang, S., et al. (2025). The PRIDE database at 20 years: 2025 update. Nucleic Acids Res. *53*, D543–D553. https://doi.org/10.1093/nar/gkae1011.

127. Sharma, V., and Majumdar, S. (2020). Comparative analysis of ChIP-exo peak-callers: impact of data quality, read duplication and binding subtypes. BMC Bioinformatics *21*, 65. https://doi.org/10.1186/s12859-020-3403-3.

128. Chu, T., Wang, Z., Chou, S.P., and Danko, C.G. (2019). Discovering Transcriptional Regulatory Elements From Run-On and Sequencing Data Using the Web-Based dREG Gateway. Curr. Protoc. Bioinformatics *66*, e70. https://doi.org/10.1002/cpbi.70.

129. Wang, Z., Chu, T., Choate, L.A., and Danko, C.G. (2019). Identification of regulatory elements from nascent transcription using dREG. Genome Res. *29*, 293–303. https://doi.org/10.1101/gr.238279.118.

130. Zhu, L.J. (2013). Integrative analysis of ChIP-chip and ChIP-seq dataset. Methods Mol. Biol. *1067*, 105–124. https://doi.org/10.1007/978-1-62703-607-8_8.

131. Wang, Q., Li, M., Wu, T., Zhan, L., Li, L., Chen, M., Xie, W., Xie, Z., Hu, E., Xu, S., et al. (2022). Exploring Epigenomic Datasets by ChIPseeker. Curr. Protoc. *2*, e585. https://doi.org/10.1002/cpz1.585.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-Actin, mouse monoclonal	Sigma-Aldrich	Cat#A2228; RRID: AB_476697
Anti-CCNL1, rabbit polyclonal	Bethyl Laboratories	Cat#:A302-058A; RRID: AB_1604263
Anti-CCNT1, rabbit polyclonal	Cell Signaling Technologies	Cat#81464; RRID: AB_2799973
Anti-CDK9, rabbit polyclonal	Cell Signaling Technologies	Cat#2316; RRID: AB_2291505
Anti-CDK11, rabbit polyclonal	Bethyl Laboratories	Cat#:A300-311A; RRID: AB_309460
Anti-DSIF, mouse monoclonal	BD Biosciences	Cat#611106; RRID AB_398419
Anti-FLAG tag, mouse monoclonal	Sigma-Aldrich	Cat#F4049; RRID: AB_439701
Anti-Lamin B1, rabbit polyclonal	Abcam	Cat#ab16048; RRID: AB_443298
Anti-Mouse IgG-HRP, rabbit polyclonal	Agilent, Dako	Cat#P02600-2; RRID: AB_2636929
Anti-Myc tag, mouse monoclonal	Cell Signaling Technologies	Cat#2276; RRID: AB_331783
Anti-Rabbit IgG-HRP, swine polyclonal	Agilent, Dako	Cat#P039901; RRID: AB_2728719
Anti-Rat IgG-HRP, rabbit polyclonal	Agilent, Dako	Cat#P0450; RRID: AB_2630354
Anti-RNA polymerase II, clone CTD 4H8, mouse monoclonal	Millipore	Cat#05-623; RRID: AB_309852
Anti-RNA polymerase II CTD repeat YSPTSPS, clone 8WG16, mouse monoclonal	Abcam	Cat#ab817; RRID: AB_306327
Anti-RNA polymerase II (RPB1) NTD, clone D8LY4, rabbit monoclonal	Cell Signaling Technologies	Cat#14958; RRID: AB_2687876
Anti-RNA Polymerase II subunit B1 (phospho-CTD Ser-2), clone 3E10, rat monoclonal	Millipore	Cat#04-1571-I; RRID: AB_10627998
Anti-RNA Polymerase II subunit B1 (phospho-CTD Ser-5), clone 3E8, rat monoclonal	Active Motif	Cat#61085; RRID: AB_2687451
Anti-RNA Polymerase II subunit B1 (phospho-CTD Ser-7), clone 4E12, rat monoclonal	Millipore	Cat#04-1570-I; RRID: AB_10618152
Anti-RNA Polymerase II subunit B1 (phospho-CTD Thr-4), clone 6D7, rat monoclonal	Active Motif	Cat#61361; RRID: AB_2750848
Anti-RNA Polymerase II subunit B1 (phospho-CTD Tyr-1), clone 3E12, rat monoclonal	Active Motif	Cat#61383; RRID: AB_2793613
Anti-RNA Polymerase II subunit B3, rabbit polyclonal	ThermoFisher Scientific	Cat#PA5-59353; RRID: AB_2645755
Anti-SPT5 (phospho Thr-806), rabbit polyclonal	Sansó et al.[101]; Robert P. Fisher, Icahn School of Medicine at Mount Sinai, New York, NY 10029, USA	N/A
Anti-a-Tubulin, clone B-5-1-2, mouse monoclonal	Sigma Aldrich	Cat#T5168; RRID: AB_477579
IRDye 800CW Anti-Mouse IgG, goat polyclonal	LI-COR	Cat#926-32210; RRID: AB_621842
IRDye 680RD Anti-Rabbit IgG, goat polyclonal	LI-COR	Cat#926-68071; RRID: AB_10956166
IRDye 680RD Anti-Rat IgG, goat polyclonal	LI-COR	Cat#926-68076; RRID: AB_10956590
Purified Mouse IgG1	Santa Cruz	Cat#sc-3877; RRID: AB_737222

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Purified Rabbit IgG	Bethyl Laboratories	Cat#P120-101; RRID: AB_479829
Bacterial and virus strains		
MAX Efficiency™ DH10Bac competent Escherichia coli	Invitrogen, Thermo Fisher Scientific	Cat#10361012
OneShot™ Stbl3™ chemically competent Escherichia coli	Invitrogen, Thermo Fisher Scientific	Cat#C737303
Chemicals, peptides, and recombinant proteins		
1-Naphtyl-PP1 (NA-PP-1)	Cayman Chemical, Sapphire Bioscience	Cat#10954
3-indole-acetic-acid	Sigma Aldrich	Cat#I3750
4-thiouridine (4sU)	Sigma Aldrich	Cat#T4509
5,6-Dichlorobenzimidazole 1-β-D-ribofuranoside (DRB)	Sigma Aldrich	Cat#1916
Alt-R S.p. HiFi Cas9 Nuclease V3, recombinant protein	Integrated DNA Technologies	Cat#1081601
AMPure XP Bead-Based Reagent	Beckman Coulter	Cat#A63880
Anne Kelso modified DMEM	Peter MacCallum Cancer Centre	N/A
Annexin-V APC	BD Biosciences	Cat#550475
AZ5576	Hashiguchi et al.[102]; AstraZeneca	N/A
BB1 RPPA blocking buffer	Zeptosens, Bayer	N/A
Benzonase	Millipore	Cat#70746
Biotin-11-CTP	Perkin – Elmer	Cat#NEL544001EA
BsmBI restriction enzyme	New England Biolabs	Cat#R0739
Buffer 3.1 restriction digestion buffer	New England Biolabs	Cat#B7203S
CDK9/CCNT1, recombinant protein	This paper	N/A
CDK11/CCNL1, recombinant protein	This paper	N/A
Chloroform	Sigma Aldrich	Cat#C2432
CLB1 RPPA lysis buffer	Zeptosens, Bayer	N/A
cOmplete Protease Inhibitors	Roche	Cat#1183580001
CSBL1 RPPA dilution buffer	Zeptosens, Bayer	N/A
CutSmart restriction digestion buffer	New England Biolabs	Cat#B7204
Dimethylformamide (DMF)	Sigma Aldrich	Cat#227056
Dimethyl sulfoxide (DMSO)	Sigma Aldrich	Cat#D8418
Doxycycline hyclate	Sigma Aldrich	Cat#D9891
DpnI	Promega	Cat#R6231
Dulbecco's Modified Eagle's medium (DMEM)	GIBCO, ThermoFisher Scientific	Cat#11995073
Dynabeads Protein A	Invitrogen, ThermoFisher Scientific	Cat#10002D
Dynabeads Protein G	Invitrogen, ThermoFisher Scientific	Cat#10003D
Dynabeads M-280 Streptavidin	Invitrogen, ThermoFisher Scientific	Cat#11206D
EcoRI-HF restriction enzyme	New England Biolabs	Cat#R3101
EcoRV-HF restriction enzyme	New England Biolabs	Cat#3195
Enhanced Chemiluminescence Plus Reagent (ECL)	Amersham, GE Healthcare	Cat#RPN2132
EZ-Link HPDP Biotin	ThermoFisher Scientific	Cat#21341
Formaldehyde Solution (37%)	Sigma Aldrich	Cat#252549
GlutaMAX™	ThermoFisher Scientific	Cat#35050061
IGEPAL CA-630	Sigma Aldrich	Cat# I8896
Lambda exonuclease	New England Biolabs	Cat#M0262
Lipopolysaccharide (LPS)	Sigma Aldrich	Cat#L2630

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
McCoy's 5A (Modified) medium	GIBCO, ThermoFisher Scientific	Cat#16600082
Odyssey Blocking Buffer	LI-COR	Cat#927-50000
OTS964	MedChem Express	Cat#HY-12467
PageRuler™ Plus Prestained Protein ladder	ThermoFisher Scientific	Cat#26619
Penicillin / Streptomycin	ThermoFisher Scientific	Cat#10378016
Phosphatase Inhibitors	Pierce, ThermoFisher Scientific	Cat#A32957
Polyethylene glycol-300 (PEG-300)	Sigma Aldrich	Cat#202371
Puromycin dihydrochloride from Streptomyces alboniger	Sigma Aldrich	Cat#P8833
RNA Polymerase II CTD repeat YSPTSPS, recombinant protein	Abcam	Cat#ab81888
Roswell Park Memorial Institute medium (RPMI)	GIBCO, ThermoFisher Scientific	Cat#11875093
Sarkosyl	Sigma Aldrich	Cat#L5125
SensiFast SYBR Hi-Rox kit	Bioline	Cat# BIO-92005
Sequa-brene	Sigma Aldrich	Cat#S2667
Schneider's Drosophila medium	GIBCO, ThermoFisher Scientific	Cat#21720024
Sf-900™ III SFM medium	GIBCO, ThermoFisher Scientific	Cat#12658019
SspI restriction enzyme	New England Biolabs	Cat#R3132
Sucrose	Sigma Aldrich	Cat#S0389
T4 DNA Ligase kit	Promega	Cat#M1801
Trypsin Gold	Promega	Cat#V5280
Triton™ X-100	Sigma Aldrich	Cat#T8787
TRIzol	Ambion, ThermoFisher Scientific	Cat#15596026
TRIzol-LS	Ambion, ThermoFisher Scientific	Cat#10296010
XhoI restriction enzyme	New England Biolabs	Cat#R0146
Critical commercial assays		
BCA Protein Assay Kit	Pierce, ThermoFisher Scientific	Cat#23227
ChIP DNA Clean and Concentrator Kit	Zymo Research	Cat#D5205
Click-iT RNA Alexa Fluor Imaging Kit	Invitrogen, ThermoFisher Scientific	Cat#C10330
CircLigase ssDNA Ligase	Astral Scientific	Cat#CL4111K
Direct-zol RNA mini-prep kit	Zymo Research	Cat#R2051
High-Capacity cDNA Reverse Transcription kit	ThermoFisher Scientific	Cat#4368814
KAPA HiFi HotStart ReadyMix	Roche	Cat#KK2602
Micro Bio-Spin™ P-30 Gel Columns, Tris Buffer	BIO-RAD	Cat#7326250
NEBNext dA-Tailing Module	New England Biolabs	Cat#E6053
NEBNext EndRepair Module	New England Biolabs	Cat#E6050
NEBNext Ultra II DNA Library Prep Kit	New England Biolabs	Cat#E7645
NEBNext Ultra II Directional RNA Library Prep Kit	New England Biolabs	Cat#E7760
phi29 DNA polymerase	New England Biolabs	Cat#M0269
Q5 High-Fidelity Mastermix	New England Biolabs	Cat#M0492
QuantSeq 30 -mRNA Seq Library Prep Kit	Lexogen	Cat#015
Quick Ligation kit	New England Biolabs	Cat#M2200S
SFCell Line 4D-Nucleofector Kit	Lonza	Cat# V4XC-2032
SG Cell Line 4D-Nucleofector Kit	Lonza	Cat# V4XC-3032
Tapestation High Sensitivity RNA kit	Agilent	Cat#067-5576

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
μMACS Streptavidin Kit	Miltenyi Biotec	Cat#130-074-101
NucleoSpin Gel and PCR Clean-up Mini kit	Macherey-Nagel	Cat#740609
NucleoSpin Plasmid Mini kit	Macherey-Nagel	Cat#740588
Deposited data		
RNA-seq data	This paper	GSE280102
ChIP-seq data	This paper	GSE280208
PRO- and TT-seq data	This paper	GSE280103
Mass spectrometry data	This paper	PXD066230
Experimental models: Cell lines		
Human: THP-1	American Type Cell Culture® (ATCC®)	Cat#TIB-202; RRID: CVCL_0006
Human: MV4;11	ATCC®	Cat#CRL-9591; RRID: CVCL_0064
Human: BJ-T	ATCC®	Cat#CRL-2522; RRID: CVCL_6573
Human: MM1.S	ATCC®	Cat#CRL-2974; RRID: CVCL_8792
Human: OCI-AML3	German Collection of Microorganisms and Cell Cultures GmbH (DSMZ)	Cat#ACC 582; RRID: CVCL_1844
Human: HEK293T	ATCC®	Cat#CRL-11268; RRID: CVCL_0063
Human: HCT116	ATCC®	Cat#CCL-247; RRID: CVCL_0291
Drosophila melanogaster: S2	ATCC®	Cat#CRL-1963; RRID: CVCL_Z232
Murine: *Mll-Af9 Nras*^{G12D} mCherry	Baker et al.,[30]; Peter MacCallum Cancer Centre	N/A
Murine: Eμ-*Myc* 4242/3.2	Shortt et al.,[103]; Peter MacCallum Cancer Centre	N/A
Murine: Eμ-*Myc* 6066/2.1	Shortt et al.,[103];Peter MacCallum Cancer Centre	N/A
Murine: Eμ-*Myc* 299/2.1	Shortt et al.,[103]; Peter MacCallum Cancer Centre	N/A
Spodoptera frugiperda: Sf21	Professor Mark Hogarth, Burnet Institute, Melbourne VIC 3000, Australia	RRID: CVCL_0518
Experimental models: Organisms/strains		
Mice: C57Bl/6	The Walter and Eliza Hall Institute, Melbourne VIC 3000, Australia	N/A
Oligonucleotides		
Scrambled (SCR) sgRNA for Human cells GCCTAGTCTCGGTAAGAGTG	Doench et al.[104]	Non-Targeting Control
sgRNA targeting Human *CDK11A/B* (for *CDK11A/B* knockout)sense: TCCCAC ATCACCGAACGATGAGAGantisense: AAACCTCTCATCGTTCGGTGATGT	Doench et al.[104]	ID_2782
sgRNA targeting Human *CDK11B* (for *CDK11B*^{G579S} homology-directed-repair) GGGTGACTTCGGGCTGGCGC	This paper	N/A
DNA donor template for Human *CDK11B*^{G579S} mutationAGCCACGCCGGCATCCTCAAG GTGAGCCCCCCTCCGAGTGGCCCATC CCAGGGAGACCTGCCGGGGCCCACTC ACAGCCGCCCATCTGTCGTTGCAGGTG AGTGATTTTGGACTGGCGCGGGAGTAC GGATCCCCTCTGAAGGCCTACACCCCG GTCGTGGTGACCCTGTGGTACCGCGCC CCAGAGCTGCTGCTTGGTG	This paper	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Oligonucleotide primer for *CDK11B* sgRNA resistance site-directed-mutagenesis Forward: CCGAACGATGAGAGAAGACTACAGCGA CAAAG Reverse: CTTTGTCGCTGTAGTCTTCTCT CATCGTTCGG	This paper	N/A
Oligonucleotide primer for *CDK11B*^{M503G} site-directed-mutagenesis Forward: GACAAGATCTACATCGTGGGG AACTATGTGGAGCACGAC Reverse: GTCGTGCTCCACATAGTTCCCCA CGATGTAGATCTTGTC	This paper	N/A
Oligonucleotide primer for *CDK11B*$^{L453I\_R455K}$ site-directed-mutagenesis Forward: CAGATGAAATTGTGGCTATAAAGAAGC TGAAGATGGAGAAGG Reverse: CCTTCTCCATCTTCAGCTTCTTTATAGCC ACAATTTCATCTG	This paper	N/A
Additional oligonucleotide sequences	This paper	Table S1
ChIP-nexus adapter and primer oligonucleotide sequences	He et al.[84]	Table S2
Recombinant DNA		
FUCas9Cherry expression vector	Aubrey et al.[105]	Addgene Cat#70182
FgH1tUTG sgRNA expression vector	Aubrey et al.[105]	Addgene Cat#70183
VSV.G viral packaging vector	Reya et al.[106]	Addgene Cat#14888
pMDLg/pRRE viral packaging vector	Dull et al.[107]	Addgene Cat#12251
pRSV-REV viral packaging vector	Dull et al.[107]	Addgene Cat#12253
MSCV IRES GFP expression vector	Tannishtha Reya	Addgene Cat#20672
pBabe Puro osTIR1-9-Myc	Lambrus et al.[108]	Addgene Cat#80074
pLX304 CCNL1	ORFeome Collaboration[109]	DNASU Cat#HsCD00442459
pMK287 mAID-Hygro	Natsume et al.[110]	Addgene Cat#72825
pFastBac Dual His6 TEV LIC 5B	Scott Gradia	Addgene Cat#30122
pLX307 CCNT1	Rosenbluh et al.[111]	Addgene Cat#98328
pFLAG CMV2 CDK9	Wang et al.[112]	Addgene Cat#28100
pCR4-TOPO MGC Human CDK11B sequence verified cDNA (BC171773)	N/A	Dharmacon MHS6278-213245351
CSII-EF-MCS mKO2-hCdt1(30/120)	Sakaue-Sawano et al.[113]; Atsushi Miyawaki, RIKEN Japan	N/A
CSII-EF-MCS mAG-hGEM(1/110)	Sakaue-Sawano et al.[113]; Atsushi Miyawaki, RIKEN Japan	N/A
Software and algorithms		
Bcl2fastq2, v2.17.1.14	https://support.illumina.com/downloads/bcl2fastq-conversion-software-v2-20.html	N/A
FASTQC, v0.11.6	http://bioinformatics.babraham.ac.uk/projects/fastqc/	N/A

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
HISAT2, v2.1.0	Kim et al.[114]	N/A
Subread, v2.0.0	Liao et al.[115]	https://subread.sourceforge.net/
Bowtie2, v2.3.4.1	Langmead and Salzberg[116]	https://bowtie-bio.sourceforge.net/bowtie2/index.shtml
Samtools, v1.9	Li et al.[117]	https://www.htslib.org/
Deeptools, v3.0.0	Ramírez et al.[118]	https://deeptools.readthedocs.io/en/develop/
MACS2, v2.2.7.1	Zhang et al.[119]	https://github.com/macs3-project/MACS
STAR, v2.5.3a	Dobin et al.[120]	https://github.com/alexdobin/STAR/
Slamdunk, v0.2.4	Neumann et al.[121]	https://t-neumann.github.io/slamdunk/
Other		
DepMap Public 20Q4v2 files (“Achilles_gene_effect.csv”, “sample_info.csv”)	DepMap[75] Meyers et al.[76] Tsherniak et al.[77]	https://depmap.org/portal/data_page/?tab=allData&releasename=DepMap+Public+20Q4+v2

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell lines

Unless otherwise specified cell lines were previously purchased from ATCC® or DSMZ and were subjected to regular mycoplasma testing through the Peter MacCallum Cancer Centre Genotyping core. THP1, MV4;11, OCI-AML3, and MM1.S cells were cultured in Roswell Park Memorial Institute medium (RPMI; ThermoFisher Scientific 11875093) supplemented with 10% (v/v) fetal bovine serum (FBS; Peter MacCallum Cancer Centre (PMCC)), 2mM GlutaMAX™ (ThermoFisher Scientific 35050061), 100U/mL penicillin and 100µg/mL streptomycin (ThermoFisher Scientific 10378016). Human embryonic kidney (HEK) 293T and hTert-immortalized BJ foreskin fibroblasts (BJ-T) cells were cultured in Dulbecco’s Modified Eagle Medium (DMEM; ThermoFisher Scientific 11995073) and HCT116 cells were cultured in McCoy’s 5A modified medium (ThermoFisher Scientific 16600082) with the same supplements. Eµ-MYC B-lymphoma cells (previously generated clones 4242/3.2, 6066/2.1 and 299/2.3[103]) were cultured in Anne Kelso-modified DMEM (PMCC) supplemented with 10% (v/v) FBS, 100U/mL penicillin and 100µg/mL streptomycin, 0.1mM L-asparagine and 50µM β-mercaptoethanol. *Drosophila melanogaster S2* cells were cultured in Schneider’s *Drosophila* medium (ThermoFisher Scientific 21720024) with the same supplements. Human and mouse cell lines were cultured at 37°C and 5 percent carbon dioxide (CO_2) and *S2* cells were cultured at room temperature and atmospheric CO_2. *Sf21* cells were a kind gift from Professor Mark Hogarth (Burnett Institute, Melbourne VIC 3000) and unless otherwise specified were cultured at room temperature in Sf-900™ III SFM (ThermoFisher Scientific 12658019).

In vivo therapy experiments

All experiments were performed with the approval of the Peter MacCallum Cancer Centre Animal Ethics Committee (AEC; 2023-11). All experiments were performed in C57Bl/6 mice purchased from the Walter and Eliza Hall Institute (Melbourne 3000 VIC, Australia) and housed in the Peter MacCallum Cancer Centre Animal Facility. OTS964 maximum-tolerated dose experiments were performed in non-tumor bearing mice; mice received OTS964 (50mg/kg or 75mg/kg) reconstituted in vehicle (10% (v/v) DMSO, 40% (v/v) PEG300, 5% (v/v) Tween80, 45% saline (0.9% v/v)) by oral gavage daily for 5 days. Mice were weighed daily and were humanely sacrificed upon loss of 20% of pre-experiment body weight. *Mll-Af9 Nras^{G12D}* MSCV-IRES-Cherry leukemia and Eµ-MYC 4242/3.2 MSCV-GFP B-lymphoma cells were previously generated.[30,122] C57Bl/6 mice were purchased from the Walter and Eliza Hall Institute (Melbourne 3000 VIC, Australia) and housed in the Peter MacCallum Cancer Centre Animal Facility. $1x10^6$ *Mll-Af9 Nras^{G12D}* MSCV-IRES-Cherry leukemia cells or $2x10^5$ Eµ-MYC B-lymphoma cells, resuspended in PBS-/-, were intravenously transplanted into mice via tail-vein injection, and mice were left for 11 days to develop circulating disease. Mice were randomly assigned to treatment groups and received either vehicle or OTS964 (50mg/kg) by oral gavage. At 24 hours post-treatment mice were culled following collection of peripheral blood from tail veins. Peripheral blood was diluted in PBS-/- (10µL blood / 250µL PBS-/-) for whole blood analysis using the Sysmex cell sorter. White blood cells were isolated from remaining peripheral blood, spleens, and tibia/femur bone marrow using red blood cell lysis buffer (144mM NH_4Cl, 17mM Tris-HCl pH7.6) and assessed for mCherry and GFP populations using the BD Fortessa or Symphony. For Eµ-MYC B-lymphoma experiments whole cell protein lysates were generated from excess splenic white blood cells and assessed using SDS-PAGE and western blotting as described herein.

METHOD DETAILS

Reagents

OTS964 was purchased from MedChemExpress (HY-12467) and 10mM stocks prepared in DMSO were stored at -80°C. AZ'5576 was provided by AstraZeneca and 10mM stocks prepared in DMSO were stored at -20°C. 3-indole-acetic-acid (3-IAA) was purchased from Sigma Aldrich (I3750) and 100mM stocks prepared in 100 percent ethanol (EtOH) were stored at -20°C. 1-Naphtyl-PP1 (1-NA-PP1) was purchased from Cayman Chemical (10954) and 10mM stocks prepared in DMSO were stored at -20°C. 5,6-Dichlorobenzimidazole 1-β-D-ribofuranoside (DRB) was purchased from Sigma Aldrich (1916) and 75mM stocks prepared in DMSO were stored at -20°C. Lipopolysaccharide (LPS) was purchased from Sigma Aldrich (L2630) and 1mg/mL stocks prepared in phosphate-buffered-saline (PBS) were stored at -20°C.

Generation of CDK11^{p110} and CDK11^{p58} expression constructs

All oligonucleotide sequences are included in Key Resources Table and Table S1, and all PCR reactions were performed using KAPA HiFi HotStart ReadyMix (Roche KK2602) as per manufacturer's instructions.

To generate MSCV-IRES-GFP FLAG-*CDK11B* expression constructs the *CDK11B* p110 and *CDK11B* p58 open-reading-frames (ORFs) were PCR amplified from a full-length *CDK11B* sequenced verified cDNA (Dharmacon MHS6278-213245351) using primer overhangs to introduce 5' EcoRI and 3' XhoI restriction enzyme (RE) recognition sites and a 5' FLAG tag. PCR products were gel-purified using the Nucleospin Gel and PCR clean-up mini kit (Macherey Nagel 740609). Purified PCR products and MSCV-IRES-GFP plasmid DNA (Addgene 20672) were incubated overnight at 37°C with EcoRI (NEB R0101) and XhoI (NEB R0146). Gel-purified RE digest products were incubated at 3:1 molar ratio of PCR insert: vector with T4 DNA ligase (Promega M1801) overnight at 16°C. One Shot™ Stbl3™ chemically competent cells (Invitrogen™ C737303) were transformed with ligation reactions by 42°C heat shock for 90 seconds, transformed colonies were selected using ampicillin (100μg/μL), and successful ORF integration was confirmed by Sanger sequencing (Australian Genomics Research Facility, AGRF) of plasmid DNA isolated using the Nucleospin Plasmid mini kit (Macherey Nagel 740588).

Site-directed mutagenesis (SDM) of MSCV-IRES-GFP FLAG-*CDK11B* p110 expression constructs was performed i) to introduce a silent mutation to remove the PAM site for the sgRNA against *CDK11A* and *CDK11B*, and ii) to generate analog sensitive (AS) mutant kinase *CDK11* (Met-503-Gly; Leu-453-Ile; Arg-455-Lys (M503G-L453I-R455K, referred to as M503G)). A whole plasmid PCR was performed using mutagenesis primers and post-amplification PCRs were incubated overnight at 37°C with DpnI (Promega R6231) to digest methylated template DNA. Stbl3™ cells were transformed with DpnI digest reactions as described above. Successful mutagenesis was confirmed using Sanger sequencing (AGRF).

To generate FLAG-*CDK11B* p110-AID expression constructs the *CDK11B* p110 ORF was PCR amplified from MSCV-IRES-GFP FLAG-*CDK11B* p110 vector DNA (with sg*CDK11A*/*CDK11B* PAM site previously removed by SDM) using primers to introduce a 5' EcoRI recognition site and 3' homology overhangs for the 5'end of the minimal auxin-inducible-degron (mAID) ORF. The mAID ORF was PCR amplified from the pMK287 mAID-Hygro vector (Addgene 72825) using primers to introduce 5'homology overhangs for the 3'end of the *CDK11B* p110 ORF and to introduce a 3'XhoI recognition site. Gel-purified PCR products were incubated at equimolar concentrations with the NEBuilder® DNA HiFi Assembly master mix (NEB E5520) for 15 minutes at 50°C. Assembled FLAG-*CDK11B* p110-mAID was PCR amplified from NEBuilder® reactions, using forward primers for the 5'end of the FLAG-*CDK11B* p110 ORF and reverse primers for the 3'end of the mAID ORF, and fusion ORF gel-purified as described above. EcoRI/XhoI digest, DNA ligation and Stbl3™ transformation of gel-purified FLAG-*CDK11B* p110-mAID DNA and EcoRI/XhoI digested MSCV-IRES-GFP plasmid DNA was performed as described above. The pBABEpuro *Os*TIR1-9Myc plasmid was obtained from Addgene (80074).

Generation of FgH1tUTG-sgRNA constructs

A short guide RNA (sgRNAs) targeting both *CDK11A* and *CDK11B* (referred to as sg*CDK11*) and non-targeting sgRNA (SCR) with BsmBI compatible overhangs were ligated as described above into FgH1tUTG-GFP or FgH1tUTG-BFP plasmid DNA that had been digested overnight with BsmBI (NEB R0739) at 55°C overnight and gel-purified. Stbl3™ cells were transformed with ligation reactions as described above and sgRNA integration was confirmed by Sanger sequencing (AGRF).

Retroviral and lentiviral transduction

THP1 cells with stable expression of Cas9 (FUCas9Cherry) were previously established.[81] FgH1tUTG-sgRNA lentivirus was generated in HEK293T cells transfected with 6μg FgH1tUTG-sgRNA (sgSCR, sg*CDK11A/B*), 3μg pMDL/pRRE (Addgene 12251), 1.5μg pRSV-REV (Addgene 12253) and 1.8μg pVSV.G (Addgene 14888) DNA using 62μg polyethyleneimine (PEI; Sigma Aldrich 913375). Lentiviral supernatant was collected at 48 hours post-transfection and target cells were transduced with lentivirus using 4μg/mL Sequa-brene (Sigma Aldrich S2667). Transduced cells were isolated by fluorescence-activated-cell-sorting (FACS) using a BDFusion3 or were selected using Blasticidin (5-10μg/mL). MSCV-IRES-GFP, MSCV-IRES-Cherry and pBABEpuro retrovirus were generated in HEK293T cells transfected with 10μg of the relevant retroviral plasmid and 10μg pCL-Ampho retrovirus packaging vector DNA using 90μg PEI. Retroviral transductions were performed with retrovirus using RetroNectin (Takara T110A). In brief, non-tissue-culture treated 6-well plates were pre-coated for 2 hours at room temperature with 30μg/mL RetroNectin diluted in PBS without $MgCl_2$ and $CaCl_2$ (-/-; 30μg per well). Retroviral supernatant from HEK293T cells was concentrated to 1mL final volume using

Amicon concentrators and retronectin-coated 6-well plates were centrifuged with concentrated virus for 1 hour at room temperature (2000g). Target cells were centrifuged onto retroviral-coated 6-well plates for 15 minutes at room temperature (1400rpm). Transduced cells were isolated using FACS or puromycin (1μg/mL). CSII-EF-MCS lentiviral vectors containing mKO2-hCdt1(30/120) or mAG-hGEM(1/110) FUCCI reporters were gifts to Peter MacCallum Cancer Centre from Atsushi Miyawaki (RIKEN, Japan,[113]). Lentivirus was generated and THP1 cells were transduced with concentrated lentiviral supernatant using the RetroNectin-based transduction protocol described above. Co-transduced mKO2 and mAG double-positive cells (THP1 FUCCI) were selected using FACS.

Generation of CDK11-mAID and CDK11-AS systems

THP1-Cas9+FgH1tUTG-BFP-sg*CDK11*+pBABEpuro+MSCV-GFP-FLAG-CDK11^{p110}-mAID, THP1-Cas9+FgH1tUTG-BFP-sg*CDK11*-+pBABEpuro-*Os*TIR1-Myc+MSCV-GFP-FLAG-CDK11^{p110}-mAID, THP1-Cas9+FgH1tUTG-BFP-sg*CDK11*+MSCV-GFP-FLAG-CDK11^{p110}-WT, and THP1-Cas9+FgH1tUTG-BFP-sg*CDK11*+MSCV-GFP-FLAG-CDK11^{p110}-AS cells were incubated with doxycycline (1μg/mL) for 1 week to induce sg*CDK11* expression. To generate the CDK11^{p110}-mAID system in MV4;11 cells, 300pmol synthetic sgRNA targeting *CDK11A* and *CDK11B* (Synthego) was incubated with 100pmol Alt-R® S.p. HiFi Cas9 Nuclease V3 (Integrated DNA Technologies (IDT), 1081061) at room temperature for 20 minutes. 5x10^5 MV4;11+pBABEpuro+MSCV-GFP-FLAG-CDK11^{p110}-mAID or MV4;11+pBABEpuro-*Os*TIR1-Myc+MSCV-GFP-FLAG-CDK11^{p110}-mAID cells were resuspended in 20uL SF nucleofector/supplement solution from the SG Cell Lines 4D-Nucleofector™ X kit (Lonza, V4XC-3032) and mixed with the Cas9-sg*CDK11* complexes prior to nucleofection using the Amaxa 4D Nucleofector X Unit (Lonza, AAF-1002X; DJ-100 program). Single cells were sorted into 96-well plates and single colonies with knockout of endogenous *CDK11A/B* were screened by SDS/PAGE and western blot as described below.

Generation of *CDK11B*$^{G579S/+}$ mutant cells

CRISPR-Cas9 homology-directed-repair was used to introduce the Gly-579-Ser (G579S) mutation into the endogenous *CDK11B* loci in THP1 and MV4;11 cells. 300pmol synthetic sgRNA targeting *CDK11B* at the mutation site was incubated with 100pmol Alt-R® S.p. HiFi Cas9 Nuclease V3 at room temperature for 20 minutes, prior to the addition of 100pmol single-stranded donor template oligonucleotide (IDT; Key Resources Table). 5x10^5 THP1 or MV4;11 cells were resuspended in 20 μL SG or SF nucleofector solution plus supplement 1 respectively and mixed with the Cas9/sgRNA/template mixture prior to nucleofection using the Amaxa 4D Nucleofector X Unit (FF100 program for THP1; DJ100 program for MV4;11). Single cells were sorted into 96 well plates 4 days post-nucleofection and mutant colonies were screened by PCR. Successful HDR was confirmed using Sanger sequencing. Two independent *CDK11B*$^{G579S/+}$ clones were generated for both THP1 and MV4;11 cells.

Cell viability and proliferation assays

For cell death and apoptosis assays, cells were seeded at 4x10^5 cells/mL and incubated with indicated concentrations of small molecule inhibitor or 3-IAA for 48 hours. For cell death analysis propidium iodide (PI; Sigma Aldrich P4170) was added to cells at a final concentration of 1μg/mL. For apoptosis analysis cells were resuspended in calcium binding buffer (20mM HEPES pH7.4, 140mM NaCl, 5mM $CaCl_2$) containing APC-conjugated Annexin-V (1:100, BD Pharmigen 556475) and incubated at room temperature for 15 minutes. For proliferation assays, 40x10^6 cells were incubated for 20 minutes in a 37°C water bath with CellTraceViolet® or CellTraceFarRed® stain (1μL/10x10^6 cells/mL in PBS -/-; ThermoFisher Scientific C34557/C34564). A uniformly stained CellTraceViolet® or CellTraceFarRed® cell population was isolated using FACS and stained cells were seeded at 4x10^5 cells/mL and incubated with indicated concentrations of small molecule inhibitor or 3-IAA for 48 hours. Cells were stained with PI (1μg/mL) prior to analysis. For THP1 FUCCI reporter cell (mKO2-hCdt1(30/120); mAG-hGEM(1/110)) assays, mKO2 single-positive (G1) and mAG single-positive (G2-M) cells were isolated by FACS (BD Fusion3) and seeded at 0.2x10^6 cells/mL with indicated concentrations of OTS964. Cells were analysed at 0 (reference), 8, and 24 hours post-sort. All analyses were performed using the BD Fortessa or BD LSRII.

Analysis of Cancer Dependency Map data

Cancer Dependency Map (DepMap) Public 20Q4v2 files (“Achilles_gene_effect.csv”, “sample_info.csv”,[75–77]) were downloaded from the DepMap portal (https://depmap.org/portal/) and analysed in RStudio Pro (v4.2.0). Data was filtered for blood cancer cell lines (lineage == “blood”), *CDK11* and *PBK*, and boxplots were generated using GraphPad Prism (v.10.0.3).

SDS-PAGE and western blotting

All protein lysates were generated from 1-2x10^6 cells. Whole cell protein lysates were generated using Laemmli buffer (2% SDS, 10% glycerol, 60mM Tris-HCl pH6.8, incubated at 95°C for 10 minutes) and nuclear protein lysates were generated using sequential lysis with i) nuclear isolation buffer (10mM HEPES, 10mM KCl, 1.5mM $MgCl_2$, 0.05% NP-40), incubated on ice for 10 minutes) and; ii) RIPA buffer (50mM Tris-HCl pH7.4, 150mM NaCl, 1% NP-40, 0.5% sodium deoxycholate, 0.1% SDS, 1mM EDTA, incubated on ice for 20 minutes). Protein lysates equivalent to equal cell number or equal protein concentration were separated using sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) using Mini-PROTEAN® TGX™ 4-20% pre-cast gels (Biorad 4561094) or lab prepared 8% or 10% polyacrylamide-SDS gels. Protein was transferred to Immobilon®-FL PVDF membrane (Millipore IPFL00010) at 4°C for 2 hours at 200mA and membranes were blocked at room temperature with 5% skim-milk diluted in

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1x Tris-buffered-saline supplemented with 0.1 percent Tween-20 (TBST) or Intercept® blocking buffer (Li-cor 927-60001). Membranes were incubated overnight with primary antibodies diluted in TBST-skim-milk or Intercept® blocking buffer supplemented with 0.1% Tween-20. Membranes were washed with TBST and incubated for 1 hour at room temperature with secondary antibodies diluted in TBST-skim-milk (HRP-conjugated) or Intercept® blocking buffer supplemented with 0.1 percent Tween-20 and 0.01 percent SDS (Li-cor fluorophore-conjugated). Membranes were washed with TBST and western blot signals were developed using enhanced chemiluminescence substrate (ECL Plus™, GE Healthcare RPN2132) and either X-ray film or the InvitrogenTM iBRIGHT™ imager for HRP-conjugated secondaries, and the Li-cor Odyssey CLx for fluorophore-conjugated secondaries. All antibodies are listed in Key Resources Table. PageRuler™ Plus Prestained Protein Ladder was used as the molecular weight marker for all gels (ThermoFisher Scientific 26619).

SLAM sequencing

$2x10^6$ cells (THP1 or *D.melanogaster* S2) were used per condition and an equal amount of *D.melanogaster* RNA was spiked into each human RNA sample to serve as an external reference (5 percent). Cells were seeded at $1x10^6$ cells/mL and incubated with 100 percent EtOH or 500μM 3-IAA for 4 hours. Cells were incubated with 100μM 4-thiouridine (4sU; Sigma Aldrich T4509) for the final hour prior to lysis in TRIzol™ reagent (ThermoFisher Scientific 15596026) and RNA isolation was performed under non-denaturing conditions. An additional sample generated from cells that had not been incubated with 4sU was used as a control. TRIzol™ samples were vortexed with 1/5th volume 100 percent chloroform prior to centrifugation at 4°C (16000g, 15 minutes) and the RNA-containing aqueous phase was decanted to a RNase/DNase-free tube. RNA was precipitated through the addition of 1 volume 100 percent isopropanol, 10mM dithiothreitol and 1μL GlycoBlue reagent (Ambion, AM9515); samples were vortexed vigorously, incubated at room temperature for 10 minutes and centrifuged at 4°C (16000g, 20 minutes). RNA pellets were washed once with 75 percent ethanol supplemented with 100μM DTT and centrifuged at room temperature (7500g for 5 minutes). RNA pellets were dried at room temperature prior to resuspension and denaturation (55°C for 10 minutes) in RNase/DNAse free water supplemented with 1mM DTT. Iodoacetamide conversion of 4sU-labelled bases was performed. 10μg human and 500ng *D. melanogaster* RNA was incubated in a 50μL reaction with 10mM iodoacetamide, 50mM $NaPO_4$ pH8.05 and 50 percent DMSO at 55°C for 15 minutes prior to the addition of DTT to 20μM. RNA was precipitated from IAA conversion reactions at -80°C (30 minutes) using NaOAc (5μL of 3M pH5.2 stock), 100 percent ethanol (125μL) and 1μL GlycoBlue reagent. RNA precipitation reactions were centrifuged at 4°C (16000g, 30 minutes) and RNA pellets were washed twice using 75 percent ethanol followed by centrifugation at 4°C (16000g, 10 minutes). RNA pellets were dried at room temperature and resuspended in RNase/DNase-free water. RNA clean-up was performed through the addition of 300μL TRIzol® reagent and purification using the Direct-zol RNA miniprep kit as per manufacturer's instructions (R2052). RNA quantity and quality was confirmed using the Agilent RNA Screentape and sample buffer (5067-5576, 5067-5577) and the Agilent 2200 TapeStation (G2964AA). Sequencing libraries were prepared using the Lexogen QuantSeq 3'mRNA-seq Library Prep Kit FWD for Illumina and single-end 75 base pair reads were sequenced using the NextSeq500.

Bulk RNA sequencing

$2x10^6$ THP1 or MV4;11 cells were used per condition. Cells were seeded at $1x10^6$ cells/mL and incubated with OTS964 and/or LPS for the indicated time points. Cells were lysed in 300μL TRIzol® reagent and RNA isolation was performed using the Direct-zol RNA miniprep kit as per the manufacturer's instructions (R2052). Sequencing libraries were prepared from 500ng RNA template using the QuantSeq 3'mRNA-Seq Library Prep kit for Illumina (Lexogen) and 75 base pair single-end reads were sequenced using the NextSeq500.

Transient-transcriptome sequencing

$50x10^6$ THP1 or *D.melanogaster* S2 cells were used per condition. Cells were seeded at $2x10^6$ cells/mL and incubated with DMSO or OTS964 for 30 minutes. Cells were pulse-labelled with 1mM 4sU for the final five minutes of the treatment period prior to lysis in TRIzol™ reagent ($50x10^6$ cells/5mL). TRIzol™ samples were vortexed with 1/5th volume 100 percent chloroform and centrifuged at 4°C (14000g, 15 minutes). The aqueous phase was decanted to a RNase/DNase-free tube and vortexed vigorously with 2.5X volume of 100 percent isopropanol prior to i) incubation at room temperature for 10 minutes, and ii) centrifugation at 4°C (14000g, 20 minutes). RNA pellets were washed once with 75 percent ethanol (diluted in RNase/DNase-free water), air-dried at room temperature and resuspended and denatured (65°C for 15 minutes) in RNase/DNase-free water. 150μg human and 15μg *D.melanogaster* RNA was combined and the sample volume was normalised to 100uL using RNase/DNase-free water. RNA samples were sonicated (Covaris S220) at maximum power for 15 seconds in 6x16mm microTube AFA Fiber Covaris tubes (520096) and successful RNA fragmentation was confirmed using the Agilent RNA Screentape and sample buffer and the Agilent 2200 TapeStation. Thiol-specific biotinylation reactions (1.5mL final volume) comprising fragmented RNA (165μg) and 300μg EZ-link HPDP-Biotin (ThermoFisher Scientific 21341), 10mM Tris pH7.4, 1mM EDTA and 40μg/mL dimethylformamide (from 200μg/mL stock, Sigma Aldrich 227056) were incubated at room temperature with constant rotation for 90 minutes. Biotinylation reactions were diluted with 1 volume 100 percent chloroform, vortexed vigorously, incubated at room temperature for 2 minutes and centrifuged at 4°C (14000g for 5 minutes). The aqueous phase was decanted to new RNase/DNase-free tubes, diluted with 1 volume 100 percent chloroform, vortexed vigorously, incubated at room temperature for 2 minutes and centrifuged at 4°C (14000g for 5 minutes). This process was repeated once. RNA was precipitated from the aqueous phase using 1 volume 100 percent isopropanol and 1/10th volume 5M NaCl; samples were vortexed vigorously prior to centrifugation at 4°C (20000g for 20 minutes). RNA pellets were washed in 75 percent ethanol (diluted in RNase/DNase-free water) and resuspended and denatured (65°C for 15 minutes) in RNase/DNase-free water. Biotinylated RNA was enriched using uMACs streptavidin beads (constant rotation at room temperature for 15 minutes) and uMACs columns (Miltenyi Biotec, 130-074-101). Columns were prepared by pre-washing once with room temperature wash buffer (100mM Tris-HCl pH7.4, 10mM EDTA, 1M NaCl, 0.1% Tween20) prior to the application of the RNA/streptavidin bead sample. Columns were washed 5 times each with i) wash buffer pre-heated to 65°C, and ii) room temperature wash buffer (900µL wash buffer volume). Enriched RNA was eluted from the column using 2 x 100µL volumes of 100mM DTT and eluates were collected into 700µL RLT buffer from the RNeasy MinElute Clean-up kit (Qiagen 74204). Enriched RNA was purified using the RNeasy MinElute kit as per manufacturer's instructions and sample quantity/quality was confirmed using the Agilent 2200 TapeStation and the High Sensitivity RNA ScreenTape and sample buffer (Agilent, 5067-5579 and 5067-5580). Sequencing libraries were prepared with the NEBNext Ultra II Directional RNA Library Prep kit (with ribodepletion, NEBNext rRNA Depletion kit, E6310) and single-end 75 base pair sequencing was performed using the NextSeq500.

Chromatin immunoprecipitation sequencing

25-50x10^6 cells were used per ChIP. Cells were seeded at 1x10^6 cells/mL and incubated with small molecule inhibitors or 3-IAA for indicated time points. Cells were washed once with PBS and cross-linked with 11% formaldehyde buffer (50mM HEPES-KOH, 100mM NaCl, 1mM EDTA) at room temperature for 30 minutes and cross-linking reactions were quenched with 125mM Glycine. Cells were washed once with ice-cold PBS prior to undergoing three rounds (10 minutes each) of nuclear lysis on ice (20mM Tris-HCl, 10mM NaCl, 2mM EDTA, 0.55 IGEPAL CA-630, supplemented with cOmplete Protease Inhibitor Cocktail (Roche 04693116001)). Isolated nuclei were resuspended in sonication buffer (20mM Tris-HCl, 150mM NaCl, 2mM EDTA, 1% IGEPAL, 0.3% SDS; 1mL / 25x10^6 cells) and sonicated at maximum power for 18 x 1 minute cycles using the Covaris S220-Focused ultrasonicator. Cleared supernatants from sonicates were diluted (20mM Tris-HCl, 150mM NaCl, 2mM EDTA, 1% Triton X-100; 1 volume) and diluted samples were incubated overnight at 4°C with a 1:1 mixture of Dynabeads Protein A/G and primary antibodies (5µg). Chromatin from *S2* cells was prepared using the same protocol and was spiked into ChIP reactions at a 1:20 ratio (5% spike-in). Protein A/G beads were washed once each in buffer 1 (20mM Tris-HCl, 150mM NaCl, 2mM EDTA, 1% Triton X-100, 0.15% SDS), buffer 2 (20mM Tris-HCl, 500mM NaCl, 2mM EDTA, 1% Triton X-100, 0.1% SDS) and buffer 3 (20mM Tris-HCl, 250mM LiCl, 2mM EDTA, 0.5% IGEPAL, 0.1% SDS, 0.5% sodium deoxycholate) and twice in Tris-EDTA buffer (10mM Tris-HCl, 1mM EDTA). Beads were re-suspended in reverse cross-linking buffer (1% SDS, 100mM $NaHCO_3$, 200mM NaCl, 30µg proteinase K) and incubated for 1 hour at 55°C. Supernatants were isolated and incubated overnight at 65°C. DNA was purified using the Zymo Research ChIP DNA Clean and Concentrator kit and sequencing libraries were prepared by the PMCC Molecular Genomics Core using the NEBNext Ultra II DNA for Illumina kit (NEB E7645). 75 base pair single-end sequencing was performed using the Illumina NextSeq 500.

Reverse phase protein array

Samples were prepared in biological triplicate and 5x10^6 cells were used per sample. THP1 cells were seeded at 1x10^6 cells/mL and incubated with DMSO or OTS964 (156nM) for 2 hours (parental, *CDK11B*$^{G579S/+}$) or with EtOH or 3-IAA (500µM) for 4 hours (CDK11-mAID). Cells were washed once with cold PBS-/- and lysed with 100µL CLB1 lysis buffer (Zeptosens, Bayer) at room temperature for 30 minutes with frequent vortexing. Lysates were centrifuged at room temperature (20000g, 5 minutes) and lysate supernatants were isolated. Protein concentration was determined using the Pierce™ BCA Protein Assay Kit (ThermoFisher Scientific, 23227) and lysates were diluted with a 9:1 mix of CSBL1:CLB1 buffer (Zeptosen). Two technical replicates of each diluted lysate were spotted onto ZeptoChips (Zeptosen) using a Nano-plotter-NP2 non-contact microarray systems (GeSim). Chips were incubated with i) BB1 blocking buffer for 1 hour (Zeptosens), ii) primary antibodies for 2 hours (1:500 dilution) and iii) Alexa Fluor® conjugated secondary antibodies for 4 hours (1:10000 dilution). Alexa Fluor® signal intensity was measured using a ZeptoREADER and ZeptoVIEW software was used to calculate relative-fluorescent-intensities normalised to Alexa Fluor® secondary antibody only control samples.

Recombinant CDK11/CDK9 generation and *in vitro* kinase assays

Full length *CDK11B*, *CDK9* (Addgene 28100), *CCNL1* (DNASU HsCD00442459), *CCNT1* (Addgene 93828) ORFs were PCR amplified using primers with ligation-independent-cloning (LIC) compatible overhangs (LIC site 1 (LIC1) overhangs for *CDK11B* and *CDK9*, LIC site 2 (LIC2) overhangs for *CCNL1* and *CCNT1*). To perform LIC, pFastBac Dual His6 TEV LIC (Addgene 30122) was linearised with SspI (LIC1, NEB R3132) or EcoRV (LIC2, NEB 3195[29]), and gel-purified linear plasmid was incubated with T4 DNA polymerase (3U; NEB M0203), 2.5mM dGTP (LIC1) or 2.5mM dCTP (LIC2), 5mM DTT, and 10µg BSA at room temperature for 30 minutes. T4 DNA polymerase treated plasmid and gel-purified PCR DNA products with LIC site-compatible overhangs were annealed at room temperature for 5 minutes, prior to incubation with 5mM EDTA at room temperature for an additional 5 minutes. Annealed reactions were transformed into OneShot™ Stbl3™ cells as described above and successful ORF integration was confirmed by Sanger sequencing (AGRF). To generate recombinant bacmid DNA, MAX Efficiency™ DH10Bac competent cells were transformed with pFastBac-*CDK11B-CCNL1* or pFastBac-*CDK9-CCNT1* DNA (30 minute incubation on ice, 45 second heat shock at 42°C, 4 hours recovery with SOC medium at 37°C, 225rpm). Transformed cells were selected using LB agar plates supplemented with kanamycin (50µg/mL), gentamicin (7µg/mL), tetracycline (10µg/mL), Bluo-gal (100µg/mL) and IPTG (40µg/mL) for 24 hours at 37°C. Positive colonies were cultured overnight at 37°C shaking at 250rpm in LB medium supplemented with recombinant bacmids were kanamycin, gentamycin, and tetracycline, and recombinant bacmid DNA was purified using the PureLink™ HiPure Plasmid DNA Miniprep kit (ThermoFisher Scientific K2100) and recombinant bacmid DNA was verified by PCR using pUC/M13 primers (ThermoFisher Scientific). To produce baculovirus, recombinant bacmid DNA was incubated with Cellfectin™ II (ThermoFisher Scientific 10362100) in Sf-900™ III SFM media for 30 minutes at room temperature, prior to transfection of $1x10^6$ log-phase *Sf21* cells cultured in Sf-900™ III SFM. *Sf21* cells were incubated for 5 hours at room temperature prior to removal of transfection solution and incubation in Sf-900™ III SFM for 3 days at 27°C. Supernatants containing P0 baculovirus were isolated, and high-titre P3 baculovirus stocks were generated by serial baculovirus amplification through repeated infection of expanded log-phase *Sf21* cell cultures. To purify recombinant CDK/CCN complexes, 50mL cultures of *Sf21* cells were infected with P3 baculovirus and incubated at for 5 days at 27°C shaking at 160rpm. Infected *Sf21* cells were pelleted, lysed on rolling platform for 5 minutes at 4°C (10mL per pellet; 10mM Tris-HCl pH 7.5, 10mM NaCl, 2mM β-ME, 0.5mM EDTA, 0.2% (v/v) IGEPAL) prior to quenching with 1.1mL NaCl (5M) and 55.5μL imidazole (1M). Lysates were centrifuged at 28000rpm in the Beckman Coulter Optima XPN-100 for 30 minutes in Thinwall Polypropylene tubes (Beckman Coulter 331372) using a SW 41 Ti swinging bucket rotor. Recombinant CDK/CCN complexes were isolated from clarified supernatants by incubation with His60 Ni Superflow resin (Takara 635677) for 1 hour at 4°C. Beads were washed 3 times with equilibration buffer (10mM Tris-HCl pH 7.5, 500mM NaCl, 5mM Imidazole, 10% (v/v) glycerol) and elution of CDK9/CCNT1 complex was performed using Talon Disposable Gravity columns (Takara 635606) with increasing concentrations of imidazole (250mM, 500mM, 1M, 2M). CDK11/CCNL1 was not eluted from beads at up to 8M imidazole and *in vitro* kinase reactions were performed using CDK11/CCNL1-bead mix. *In vitro* kinase assays were performed as described previously.[81] Purified CDK9/CCNT1 (eluted) or CDK11/CCNL1 (on-bead) complex was incubated with 0.5μg recombinant human RNA polymerase II CTD repeat TSPTSPS peptide (Abcam ab81888) in kinase reaction buffer (25mM Tris-HCl pH7.4, 2mM DTT, 10mM $MgCl_2$) supplemented with 200mM Mg^+-ATP (Sigma Aldrich A9187) for 1 hour at 30°C (300rpm). Reactions were quenched by the addition of 6X laemmli lysis buffer and incubation at 95°C and were analysed by SDS-PAGE and western blotting as described previously.

Immunoprecipitation mass spectrometry

$30x10^6$ THP1 cells were used for each IP in triplicate and were seeded at 1x10̂6 cells/mL and incubated with small molecule inhibitors for 2 hours. Cells were washed with 10mL cold PBS and subjected to sequential lysis on ice with 2X pellet volumes of hypotonic lysis buffer (10mM HEPES pH7.5, 10mM KCl, 150mM $MgCl_2$, 40 minutes), nucleoplasmic lysis buffer (20mM HEPES pH7.9, 1.5mM $MgCL_2$, 150mM KAc, 10% glycerol, 0.05% IGEPAL, 20 minutes), low salt chromatin digestion buffer (20mM HEPES pH7.9, 3mM EDTA, 1.5mM $MgCl_2$, 150mM NaCl, 10% glycerol, 0.1% IGEPAL, 250U/mL Benzonase, 1 hour) and high salt chromatin digestion buffer (20mM HEPES pH7.9, 3mM EDTA, 1.5mM $MgCl_2$, 500mM NaCl, 10% glycerol, 0.1% IGEPAL, 20 minutes) with samples centrifuged (20000g) for 20 minutes at 4°C after each incubation. High salt chromatin samples were equilibrated with 6X pellet volume dilution buffer (20mM HEPES pH7.9, 3mM EDTA, 1.5mM $MgCl_2$, 10% glycerol, 0.1% IGEPAL) prior to centrifugation. Nucleoplasmic, low salt and high salt fractions were pooled and tumbled overnight at 4°C with 5μg antibody or IgG isotype-control and 25μL Dynabeads Protein A (rabbit IgG or CDK11) or Protein G (mouse IgG or Pol II). Beads were washed with wash buffer (20mM Tris-HCl pH7.5, 150mM NaCl, 1.5mM $MgCl_2$, 3mM EDTA, 10% glycerol, 0.1% IGEPAL) 5 times and with 100mM ammonium bicarbonate twice. Beads were resuspended in 100mM AMBIC and incubated overnight at 37°C with mass-spectrometry grade Trypsin Gold (400ng, Promega V5280). Samples were supplemented with additional trypsin (400ng) and incubated at 37°C for 4 hours. Peptide-containing supernatants were acidified with formic acid (3% final concentration) and isolated using Pierce™ C18 Spin tips that were pre-conditioned with acetonitrile (100%, 80%, 80%) and formic acid (0.5%). Peptide loaded C18 Spin tips were washed twice (formic acid 0.1%), peptides were eluted in acetonitrile/formic acid (80% acetonitrile in 0.1% formic acid) and excess acetonitrile was removed using a SpeedVac™.

LC-MS/MS of purified peptides was performed at the Bio21 Mass Spectrometry and Proteomics Facility (Parkville, VIC 3052, Australia). Samples were analysed using the QExactive plus Orbitrap mass spectrometer (Thermo Scientific) with a nanoESI interface in conjunction with an Ultimate 3000 RSLC nanoHPLC (Dionex Ultimate 3000). The LC system was equipped with an Acclaim Pepmap nano-trap column (Dinoex-C18, 0 Å, 75 μm × 2 cm) and an Acclaim Pepmap RSLC analytical column (Dinoex-C18, 100 Å, 75 μm × 50 cm). The tryptic peptides were injected to the enrichment column at an isocratic flow of 5 μL/min of 2% v/v CH3CN containing 0.1% v/v formic acid for 5 min applied before the enrichment column was switched in-line with the analytical column. The eluents were 0.1% formic acid (solvent A) and 100% CH3CN in 0.1% formic acid (solvent B). The flow gradient was (i) 0–6min at 3% B, (ii) 6–40 min, 3–25% B (iii) 40–48 min, 25–45% B (iv) 48–50 min, 40–80% B (v) 50–53 min, 85–85% B (vi) 53–54 min 85–3% and equilibrated at 3% B for 10 minutes before the next sample injection. The QExactive plus mass spectrometer was operated in the data-dependent mode, whereby full MS1 spectra were acquired in positive mode, 70000 resolution, AGC target of 3e6 and maximum IT time of 50ms. Fifteen of the most intense peptide ions with charge states ≥2 and intensity threshold of 1.7e4 were isolated for MS/MS. The isolation window was set at 1.2m/z and precursors fragmented using normalized collision energy of 30, 17 500 resolution, AGC target of 1e5 and maximum IT time of 100ms. Dynamic exclusion was set to be 30 seconds.

MaxQuant was used for label free quantitation of detected peptides and spectral counts were used to characterise protein interactomes across different conditions.[123] Peptides were filtered to remove those that were potential contaminants, part of a protein derived from the reversed part of the decoy database or with an Andromeda score of < 40. Peptide enrichment comparisons were performed using the stat_test() function of the IPinquiry4 R package (div="DESeq") to identify proteins that were significantly enriched ($p<0.05$) in CDK11 and Pol II IPs compared to matched IgG controls,[124] AnnotationDbi was used to filter for nuclear proteins (GO:0005634), and interaction networks were generated using STRING.[125] The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE[126] partner repository with the dataset identifier PXD066230.

Co-immunoprecipitation

For each IP, 10×10^6 THP1 cells seeded at 1×10^6 cells/mL were treated with OTS964 for 2 hours. Cells were subjected to sequential lysis as per IP-MS samples and pooled nucleoplasmic, low salt, and high salt chromatin fractions were tumbled for 1 hour at 4°C with 5μg antibody or purified rabbit IgG and 25μL Dynabeads Protein A. Beads were washed 5 times and eluted by incubation at 95°C with concentrated Laemmli loading buffer (4% SDS, 20% glycerol, 120mM Tris-HCl pH6.8, 10% β-mercaptoethanol, bromophenol blue). SDS-PAGE and western blotting of IP eluate and pooled input samples was performed as above.

ChIP-nexus

Protocol was adapted from.[84] 25×10^6 THP1 cells were used per treatment condition. Cells were seeded at 1×10^6 cells/mL and incubated with small molecule inhibitors for 2 hours. Formaldehyde cross-linking, glycine quenching, nuclear isolation, sonication, and chromatin incubation with antibody and Dynabeads Protein A/G (50μL slurry per IP, equivalent to 5μL packed volume) was performed as described for ChIP-seq. Post-enrichment Dynabeads were washed one time each with 1mL of cold wash buffer A (10mM Tris pH8, 1mM EDTA, 0.1% (v/v) TritonX-100), B (20mM Tris pH8, 5mM EDTA, 150mM NaCl, 1% (v/v) TritonX-100, 5.2% (w/v) sucrose, 0.2% (w/v) SDS), C (5mM Tris pH8, 0.5mM EDTA, 250mM NaCl, 0.5% (v/v) TritonX-100, 25mM HEPES, 0.05% (w/v) sodium deoxycholate), D (10mM Tris pH8, 250mM LiCl, 10mM EDTA, 0.5% (v/v) IGEPAL, 0.5% (w/v) sodium deoxycholate), and 10mM Tris (pH8), 1mL per wash. All wash buffers were supplemented with cOmplete Protease Inhibitor Cocktail. On-bead end-repair enzymatic reaction was performed using the NEBNext End Repair kit (E6050S; 5μL beads, 1μL enzyme mix, 5μL reaction buffer, 45μL ultrapure water) at 25°C for 30 minutes. Beads were washed one time each with 1mL of cold wash buffer A, B, C, D and 10mM Tris. On-bead dA-tailing was performed using the NEBNext dA-tailing kit (E6053S; 5μL beads, 1μL Klenow fragment, 5μL reaction buffer, 39μL ultrapure water) at 37°C for 30 minutes, and beads were washed one time each with 1mL of cold wash buffer A, B, C, D and 10mM Tris. On-bead ligation of ChIP-nexus adapters was performed using the NEB Quick Ligation kit (M2200S; 5μL beads, 3μL 1μM ChIP-nexus adapters (see Table S2[84]), 3μL Quick Ligase, 12.5μL reaction buffer, 3.5μL ultrapure water) at 25°C for 30 minutes, and beads were washed one time each with 1mL of cold wash buffer A, B, C, D and 10mM Tris. On-bead adapter barcode extension was performed using the NEB phi29 DNA polymerase kit (M0269S; 5μL beads, 1μL phi29 DNA polymerase, 5μL reaction buffer, 1μL 10mM dNTPs, 0.5μL BSA (20mg/mL), 37.5μL ultrapure water) at 37°C for 30 minutes, and beads were washed one time each with 1mL of cold wash buffer A, B, C, D and 10mM Tris. On-bead exonuclease digestion was performed using the NEB Lambda exonuclease kit (M0262S; 5μL beads, 4μL Lambda exonuclease, 10μL reaction buffer, 1μL TritonX-100 (10% v/v stock), 5μL DMSO, 75μL ultrapure water) at 37°C for 1 hour with 1000rpm, and beads were washed 3 times with 1mL cold RIPA buffer (50mM HEPES pH7.5, 500mM LiCl, 1mM EDTA, 0.7% (w/v) sodium deoxycholate, 1% (v/v) IGEPAL). Beads were incubated overnight at 65°C with Proteinase K (20μg/mL in 25mM Tris pH 8, 5mM EDTA, 0.5% (w/v) SDS). To isolate DNA, 1 volume phenol:chloroform:IAA (25:24:1) was added and samples were vortexed for 10 seconds, incubated at room temperature for 3 minutes and centrifuged at 20000g for 5 minutes at 4°C. The aqueous phase isolated and DNA was precipitated by adding 3μL GlycoBlue reagent, 12μL 5M NaCl and 750μL cold ethanol. Samples were vortexed for 1 minute and incubated at -80°C for 30 minutes. Samples were centrifuged at 20000g for 30 minutes at 4°C and DNA pellet was washed with 500μL 70% ethanol. Washed DNA pellets were air-dried prior to resuspension in 12μL ultrapure water. Purified DNA was denatured at 80°C for 5 minutes prior to single-stranded DNA circularization was performed using CircLigase ssDNA Ligase (Lucigen CL4111K; 12μL purified DNA, 0.5μL CircLigase ssDNA ligase, 1.5μL reaction buffer, 0.75μL 50mM $MnCl_2$, 0.75μL 1mM ATP) at 80°C for 1 hour. Library PCR was performed using the NEB Q5 High-Fidelity DNA polymerase mastermix (M0492S; 15μL CircLigase reaction, 25μL Q5 DNA polymerase 2X mastermix, 1μL 10μM Nexus universal primer, 1μL 10μM Nexus index primer, 34μL ultrapure water). Primer sequences are included in Table S2.[84] PCRs were purified using AMPure XP beads where a 1:2 ratio of PCR:beads were incubated at room temperature for 5 minutes prior to washing of beads with 70% ethanol. Libraries were eluted using ultrapure water and analyzed using Agilent D1000 screentape assay and TapeStation 4150. Single-end 75 or 100 base pair sequencing was performed using the NextSeq500 or NextSeq2000.

PRO-seq

10×10^6 THP1 cells were used per treatment condition and cells were seeded at 1×10^6 cells/mL on the day of the experiment. For DRB-release assays, cells were seeded at 1×10^6 cells/mL and incubated with 100μM DRB for 4 hours. DRB-treated cells were washed twice with 10mL cold PBS and resuspended in 10mL pre-warmed media containing OTS964 (156nM), AZ'5576 (170nM) or equivalent DMSO prior to incubation at 37°C for 10 minutes (release). For the t0 (no release) control, DRB-treated cells were incubated on ice. Following release, cells were washed once with 10mL cold PBS prior to 3 rounds (1 x 10mL, 5 min on ice; 2 x 5mL, 1 min on ice) of incubation with nuclear isolation buffer (10mM Tris-HCl pH 7.4, 300mM sucrose, 10mM KCl, 5mM $MgCl_2$, 1mM EGTA, 0.05% (v/v) Tween20, 0.1% (v/v) IGEPAL, 0.5mM DTT) supplemented with cOmplete Protease Inhibitor Cocktail and RNase inhibitor (2 units/mL; Applied Biosystems N8080119). Isolated nuclei were resuspended in storage buffer (100μL per sample; 10mM Tris-HCl pH 8, 25% (v/v) glycerol, 5mM $MgCl_2$, 0.1mM EDTA, 5mM DTT) and snap-frozen. *In vitro* run-on reactions were performed as follows: 100μL nuclei were mixed with 50μL 2% (w/v) Sarkosyl (Sigma Aldrich L5125) and pre-warmed to 37°C prior to the addition of 100μL 2X biotin-CTP reaction mix (10mM Tris-HCl pH 8, 5mM $MgCl_2$, 1mM DTT, 300mM KCl, 0.05mM biotin-11-CTP (Perkin Elmer NEL544001EA), 0.5μM CTP, 0.25mM each ATP, GTP, UTP, 0.02% (v/v) RNase inhibitor); reactions were mixed by pipetting 15 times and incubated at 37°C for 3 minutes and reactions were quenched through the addition of 500μL TRIzol™ LS reagent (Invitrogen 10296028). TRIzol™ LS samples were vortexed with 130μL chloroform prior to centrifugation at 4°C (14000g, 5 minutes) and recovery of the aqueous phase to a RNase/DNase-free tube. RNA was precipitated through the addition of 2.5X volumes 100% ethanol and 1μL GlycoBlue reagent; samples were vortexed vigorously for 1 minute, incubated at room temperature for 10 minutes and centrifuged at 4°C (14000g, 20 minutes). RNA pellets were washed with 70% ethanol prior to reconstitution in 20μL RNase/DNase-free water. RNA was denatured at 65°C (40 seconds) and incubated on ice for 13 minutes with ice cold NaOH (5μL 1M, 200mM final) prior to addition of Tris-HCl pH6.8 (25μL 1M, 500mM final). Buffer exchange was performed using Micro Bio-Spin™ P-30 gel columns as per manufacturer's instructions (BioRad 7326250, 2 columns per sample). To enrich biotin-CTP-incorporating RNA, hydrolysed RNA was incubated with 30μL streptavidin DynaBeads (Invitrogen 11206D; pre-washed with 1) 100mM NaOH, 50mM NaCl and 2) 100mM NaCl and resuspended in binding buffer (100mM Tris-HCl pH 7.4, 300mM NaCl, 0.1% (v/v) TritonX-100) at room temperature for 20 minutes (tumbling at 8rpm). Beads were placed on ice and washed 3 times each (500μL per wash) with ice-cold high salt wash buffer (50mM Tris-HCl pH 7.4, 2M NaCl, 0.5% (v/v) TritonX-100), binding buffer, and low salt wash buffer (5mM Tris-HCl pH 7.4, 0.1% (v/v) TritonX-100), prior to resuspension in 300μL TRIzol™. TRIzol™ samples were vortexed with 60μL chloroform prior to centrifugation at 4°C (14000g, 5 minutes) and recovery of the aqueous phase to a RNase/DNase-free tube. This was repeated once and the combined aqueous phase was vortexed with 1 volume chloroform prior to centrifugation at 4°C (14000g, 5 minutes) and recovery of the aqueous phase to a new RNase/DNase-free tube. Biotin-enriched RNA was precipitated through the addition of 2.5X volumes 100% ethanol and 1μL GlycoBlue reagent; samples were vortexed vigorously for 1 minute, incubated at room temperature for 10 minutes and centrifuged at 4°C (14000g, 20 minutes). RNA pellets were washed with 70% ethanol prior to reconstitution in 5μL RNase/DNase-free water. Enzymatic end-repair was performed as follows: biotin-enriched RNA was incubated at 37°C for 1 hour each with RppH (NEB M0356S; 5 units prepared in 1X ThermoPol reaction buffer (NEB B9004) with 10 units RNase inhibitor) and T4 polynucleotide kinase (NEB M0201; 25 units prepared in 1X PNK buffer with 50 units RNase inhibitor and 1mM ATP). Reactions were quenched with 300μL TRIzol™ and biotin-enriched RNA was isolated using chloroform extraction and ethanol precipitation as described above. RNA pellets were washed with 70% ethanol and resuspended in 5μL RNase/DNase-free water. Sample quantity/quality was assessed using the Agilent 4150 TapeStation and the High Sensitivity RNA ScreenTape and sample buffer. Sequencing libraries were prepared with the NEBNext Ultra II Directional RNA Library Prep kit (with ribodepletion, NEBNext rRNA Depletion kit, E6310) and single-end 100 base pair sequencing was performed using the Illumina NextSeq500 or NextSeq2000.

Next-generation sequencing analysis

Sequencing files were demultiplexed using Bcl2fastq (v2.17.1.14) and quality-control was performed using FASTQC (v0.11.6). FASTQ files were trimmed using cutadapt (v2.1) and aligned to the Hg19 *Homo sapiens* or dm3 *Drosophila melanogaster* reference genome using HISAT2 (v2.1.0; 3'UTR RNA-seq, TT-seq, PRO-seq) or bowtie2 (v2.3.4.1; ChIP). SAM files were converted to BAM files using Samtools (v1.9) view function. BAM files were sorted and indexed using Samtools, and PCR duplicates were removed for ChIP samples (rmdup). For 3'UTR RNA-seq analyses SubRead FeatureCounts tool (v2.0.0) was performed to generate read counts. Differential gene expression analysis was performed in R (v.3.42.2) using Voom-Limma and heatmaps were generated using ggplot2. Metascape was used to perform gene ontology analysis was performed on genes displaying a log fold change of ≤ -1 or ≥ 1 (adjusted p.value ≤ 0.05) between OTS964 and DMSO treatment groups in both THP1 and MV4;11 cells and between 3-IAA and EtOH treatment groups in THP1 BABEpuro;*Os*TIR1-Myc; MSCV-GFP-FLAG- CDK11^{P110}-mAID cells. For ChIP, TT-seq and PRO-seq BAM files were converted to BigWig files using deeptools (v.3.0.0) using the bamCoverage tool normalising using CPM (ChIP, PRO-seq) or RPKM (TT-seq) and were scaled according to the proportion of dm3/Hg19 reads. Deeptools computeMatrix function was used to compute average read density across transcriptional units (scale-regions) or centred around the transcription start site (reference-point), with the resulting matrices used to generate chromatin occupancy profile plots and heatmaps (Deeptools plotHeatmap). Read coverage across whole transcriptional units (total), TSS (-50bp to +100bp of TSS) and gene body (+100bp of TSS to -50bp of TES) regions was plotted in RStudio using boxplot function. Peak calling was performed using MACS2 (v.2.2.7.1) using the callpeak function.

For ChIP-nexus, FASTQ files were trimmed using trimgalore (v.0.4.4), aligned to the human hg38 genome using STAR (v.2.5.3a), and peaks were called using GEM (v.3.4; –range 250 –smooth 5, -mrc 100, script.4.gem_all.sh[127]). For PRO-seq data FASTQ files were also trimmed using trimgalore (v.0.4.4) and processed with Proseq2.0 with parameters for single-end sequencing from 3' end of nascent RNA.[128] Reads were aligned to hg38 genome, and bigwig files were generated in a format compatible with dREG.[129] Strand-specific, non-normalized bigWig files were uploaded to the dREG portal, and resulting peak calls in BED format were downloaded for downstream analysis. Transcription start site (TSS) coordinates for the human genome (hg38) were obtained using the R package TxDb.Hsapiens.UCSC.hg38.knownGene. Blacklisted TSS regions from the ChIPPeakAnno package[130]were lifted over from hg19 to hg38 and excluded from further analysis. ChIP-nexus and PRO-seq peaks from all samples were overlapped with genomic regions of 500bp around each TSS and quantified using the *getTagMatrix* function from the ChIPseeker package.[131] Replicates were concatenated, and the average signal per sample was visualized using ggplot2, applying linear-model smoothing with a 25-degree polynomial fit. To assess differential signal at TSSs between samples, a 125 bp window downstream of the TSS was analyzed, retaining only those TSSs where both samples exhibited at least one peak. In cases with multiple peaks, only the peak with the highest score was considered. Statistical significance between distributions was assessed using the Wilcoxon rank-sum test.

For SLAM-seq FASTQ files were aligned to the combined Hg38/dm6 reference genome using the map dunk of the slamdunk package (v0.2.4[121]). BAM files were filtered (filter dunk) and snp variants were called using the snp dunk. T-C converted reads were counted using the count dunk using a T-C conversion threshold of ≥ 2. T-C conversion rates were determined using the Alleyoop function of the slamdunk package. Human T-C converted reads were normalised to *drosophila melanogaster* T-C converted reads for each sample.

Reverse transcription quantitative polymerase chain reaction

For RDH gene analysis THP1 cells were treated with OTS964 for 2 and 24 hours. For heat shock experiments, HCT116 cells were seeded in 6-well plates (1×10^5 cells/well) 18 hours prior to incubation with DMSO or OTS964 at 37°C or 42°C (heat shock) for 1 hour. For serum starvation and refeeding experiments, BJ-T cells were seeded in 10cm dishes (5×10^5 / dish) 18 hours prior to serum starvation (0% FBS, 0.5% BSA) for 24 hours. Cells were refeeded with culture medium containing 10 percent FBS for 4 hours and treated with DMSO or OTS964 for the final 2 hours. For both experiments, cells were scraped in 300μL TRIzol® reagent and RNA isolation was performed using the Direct-zol RNA miniprep kit as per manufacturer's instructions. cDNA was prepared from 1ug RNA template using the High-Capacity cDNA Reverse Transcription kit (ThermoFisher Scientific 4368814) in 20μL reactions incubated at 25°C (10 minutes), 37°C (2 hours) and 85°C (5 minutes). cDNA reactions were diluted (400uL final volume) using RNase/DNase free water. qPCR reactions comprising SensiFAST™ SYBR® Hi-Rox 2X mastermix (50 percent), qPCR forward and reverse oligonucleotide primers (0.5μM each, Table S1) and diluted cDNA template (40 percent) were run on the Roche LightCycler® 480 instrument. Normalization was performed to CT values obtained for the *L32* housekeeping gene and analysis was performed using the ΔΔCT method.

Click-iT® EU labeling

HCT116 cells were seeded in 6 well plates (1×10^5 cells/well) and incubated at 37°C for 18 hours prior to serum starvation (0% FBS) for 24 hours. Cells were re-fed with culture medium containing 10% FBS for 6 hours, treated with DMSO or OTS964 for the final 2 hours, and labelled with 500μL EU reagent from the Click-iT™ RNA Alexa Fluor™ Imaging Kit (ThermoFisher Scientific 10330) for the final 30 minutes. Cells were incubated overnight at room temperature (protected from light) with fix/permeabilization buffer (0.5% paraformaldehyde and 0.2% Tween20 diluted in 0.1% BSA and 0.1% azide in PBS -/-). Fixed cells were washed once with PBS -/- prior to the Click-iT™ RNA reaction. Cells were incubated at room temperature for 30 minutes with Click-iT™ RNA reaction buffer (86μL), CuSO4 (4μL), Alexa Fluor 647 (0.125μL) and Click-iT™ buffer additive (10.3μL), with all stock reagents prepared according to the Click-iT™ RNA Imaging kit instructions. Cells were washed once with Click-iT™ RNA wash buffer, three times with PBS -/- supplemented with 0.1% Tween20 and once with fix/permeabilization buffer. Cells were resuspended in un-supplemented PBS -/- and analysed using the BDFortessa.

QUANTIFICATION AND STATISTICAL ANALYSIS

The number of replicates (as the number of independent experiments for *in vitro* and cell-based assays, number of mice for *in vivo* experiments) and details of statistical tests performed and significance values are indicated in the figure legends. Data is presented as the mean +/- standard error of the mean (SEM). ANOVA and Student's *t*-Test analyses were performed using GraphPad Prism10 software (La Jolla, CA) and unpaired, two-sided *t*-tests were performed using RStudio (v.4.2.0). For all analyses a p-value of < 0.05 was considered statistically significant.